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Key Points:

- A long short-term memory (LSTM) model is built to forecast SYM-H index multiple-hour ahead using interplanetary magnetic field (IMF) measurements and SYM-H values
- Prediction uncertainties from the LSTM model are estimated and turn out to be considerable in the critical phases of geomagnetic storms
- The uncertainty quantification is found to be crucial to achieve a reliable forecasting model and determine the optimal look-forward

Supporting Information:

Supporting Information may be found in the online version of this article.

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Forecasting Geomagnetic Storm Disturbances and Their Uncertainties Using Deep Learning

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Abstract Severe space weather produced by disturbed conditions on the Sun results in harmful effects both for humans in space and in high-latitude flights, and for technological systems such as spacecraft or communications. Also, geomagnetically induced currents (GICs) flowing on long ground-based conductors, such as power networks, potentially threaten critical infrastructures on Earth. The first step in developing an alarm system against GICs is to forecast them. This is a challenging task given the highly non-linear dependencies of the response of the magnetosphere to these perturbations. In the last few years, modern machine-learning models have shown to be very good at predicting magnetic activity indices. However, such complex models are on the one hand difficult to tune, and on the other hand they are known to bring along potentially large prediction uncertainties which are generally difficult to estimate. In this work we aim at predicting the SYM-H index characterizing geomagnetic storms multiple-hour ahead, using public interplanetary magnetic field (IMF) data from the Sun-Earth L1 Lagrange point and SYM-H data. We implement a type of machine-learning model called long short-term memory (LSTM) network. Our scope is to estimate the prediction uncertainties coming from a deep-learning model in the context of forecasting the SYM-H index. These uncertainties will be essential to set reliable alarm thresholds. The resulting uncertainties turn out to be sizable at the critical stages of the geomagnetic storms. Our methodology includes as well an efficient optimization of important hyper-parameters of the LSTM network and robustness tests.

Plain Language Summary Geomagnetic storms are disturbances of the geomagnetic field caused by interactions between the solar wind and particle populations mainly in the Earth's magnetosphere. These time-varying magnetic fields induce electrical currents on long ground-based conductors that can damage power transmission grids and other critical infrastructures on Earth. As a first step to forecast the ground magnetic perturbations caused by geomagnetic storms at specific mid-latitude locations, the objective of this work is to predict the SYM-H activity index, which is generated from ground observations of the geomagnetic field at low and mid-latitudes, and which provides a measure of the strength and duration of geomagnetic storms. We use the IMF data measured by the Advanced Composition Explorer spacecraft at the L1 Lagrangian point and SYM-H values to forecast the behavior and severity of geomagnetic storms multiple-hour ahead. This forecasting is done using a type of artificial neural network model called long short-term memory. We also propose robust ways to estimate the uncertainties of these predictions, which help us to better understand machine-learning models in activity indices prediction and lead to more accurate and reliable forecasting of geomagnetic storms and their ground effects in the near future.

1. Introduction

In the last decades, our society has become more interdependent and complex than ever before and is highly dependent on relevant technological structures which can be very vulnerable to the effects of space weather (SW). The latter has its origin in the solar activity and their associated events, such as coronal mass ejections and co-rotating interaction regions. Among other effects, these phenomena have an impact on the electrical current systems surrounding the Earth, enhancing them and thus causing large magnetic field fluctuations that propagate down to the ground. The electric field associated with these fluctuations, which is influenced by the interaction with the conductive earth, induces telluric currents in the uppermost solid layers and geomagnetically induced currents (GICs) in long conductors running on the surface. These GICs may cause disturbances, interruptions, and even long-term damage to critical infrastructures such as railways, oil and gas pipelines and power grids (Boteler et al., 1998), with drastic social, economic and even political consequences. Because of this, it is necessary to forecast these hazardous events to prevent their effects. In particular, it is fundamental to develop highly reliable predictive models to build real-time early warning systems and vulnerability maps to alert on potential impact of future violent solar storms on critical infrastructures. In order to have such trustworthy forecasting models, it is crucial to estimate the associated uncertainties, especially with regard to reliability assessment and decision making.

Forecasting and modeling the behavior of the GICs is a challenging task. The intensity of the GICs is determined by the strength of the geoelectric field, but the latter measurements are rarely available. Because GICs are driven by temporal changes in the magnetic field, if we have an estimate of the resistivity structure below a specific location, variations in the magnetic field measured by ground magnetometers can in principle be used as the input parameter for deriving the GICs built up (e.g., Torta et al., 2017 for GICs in the power grid). Predicting geomagnetic indices, which attempt to condense a rich set of information about the status of the magnetosphere in a single number (Mayaud, 1980), is simpler and has always been a very attractive area for machine-learning (ML) applications (Camporeale, 2019). Although attempts to forecast geomagnetic indices started several decades ago (e.g., Burton et al., 1975), they feature highly non-linear dependencies which are not yet well understood, and their forecasting is still an open and intensive area of research. Perhaps not surprisingly, recent efforts have been exploiting the large expressiveness offered by modern ML models (e.g., Raghu et al., 2017), and their ability to characterize complicated multidimensional data sets. The present work follows such a trend by investigating advanced ML techniques to predict the behavior of geomagnetic storms.

Despite our ultimate goal is to build an alarm system against GICs and provide a vulnerability map in Spain, given the complexity of the problem, the scope of this paper is to establish the basis by presenting a simple though highly reliable forecasting model following the trends on the current literature (see Section 2) though also including the estimation of the associated uncertainties of our forecasting model. More specifically, this paper presents the prediction of the SYM-H index, at a given time in advance, using a type of artificial neural network (NN) model called long short-term memory (LSTM) (Hochreiter & Schmidhuber, 1997) especially conceived for describing, among others, non-linear time-series data. The SYM-H index describes the geomagnetic disturbances at low and middle latitudes in terms of longitudinally symmetric disturbances of the horizontal component of the geomagnetic field (Iyemori, 1990). This index is known to track very well the evolution in time, the topology and intensity of geomagnetic storms and their relation with solar source phenomena (J. Wanliss, 2005; J. Wanliss & Uritsky, 2010). To feed the LSTM network, we use time-series data from the Sun-Earth L1 Lagrange point tracking several covariates describing the interplanetary magnetic field (IMF) and its different components in addition to the SYM-H index.

Highly-parameterized NNs as the ones we use in this work (as well as in recent literature) carry an important amount of intrinsic prediction uncertainty, called among statisticians "epistemic" uncertainty. This should be taken into account, where possible, in any scientific application, even more in those with direct impact on society as the present study. Yet another issue with those models is the presence of parameters (named "hyper-parameters" in the ML community) which are not directly optimized during the fitting processes, but whose impact on the predictions are potentially very large. Consequently, an efficient Bayesian optimization strategy is presented. Furthermore, robustness in non-linear time-series predictions obtained from ML models (including LSTM) can be challenging due to their complex and often unpredictable nature. However, several techniques exist (reshuffling, etc.) and can be used to test and improve the robustness of the models.

This paper is organized as follows. Section 2 revises recent related and reference works and pointing out the main differences between these works and the work presented in this paper. Section 3 describes the data set used, the data preparation, the list of geomagnetic storms used in each sub-data set and the variables used to train the LSTM model. Section 4 details the deep-learning architecture used in this work to predict the SYM-H index, how the hyper-parameters are optimized, how the robustness and accuracy of the model is ensured, and how the prediction uncertainties are estimated. Section 5 presents and discusses the results of our LSTM model, comparing our 1- and 2-hr ahead predictions with some previous works, and we also discuss the behavior of the uncertainties for multiple-hour ahead predictions and the effects of systematics uncertainties. Finally, some conclusions are outlined in Section 6 together with some ideas for future works.

2. Related Works

Efforts to forecast geomagnetic indices (Kp, Dst, SYM-H, etc.) date back to the late 70s and early 80s decades (Burton et al., 1975; Mayaud, 1980), which started using linear prediction models which were unable to capture well enough the complexity of the response of the magnetosphere to SW. For this reason, the SW community started to rely on the arbitrarily high expressiveness of deep-learning models (e.g., Bhaskar & Vichare, 2019; Gruet et al., 2018; Siciliano et al., 2021; Tan et al., 2018).

Among the works predicting geomagnetic indices and using deep-learning models, there are a few which are highly relevant references to our work. These works provide forecasting of the SYM-H geomagnetic index, which has a finer time granularity with respect to other indices (e.g., Dst index, J. A. Wanliss & Showalter, 2006), thus being advantageous from the point of view of an alert system. The work by Siciliano et al. (2021) forecasts 1-hr ahead the SYM-H index using two different NN models: a LSTM and a convolutional neural network (CNN), the latter being typically used for image recognition tasks (Zhang, 1988; Zhang et al., 1990). While they have obtained good performances with the CNN compared to the LSTM (in some cases even slightly better), in our study we concentrate on the LSTM only, which for us delivered similar performances as the CNN. Posterior work by Collado-Villaverde et al. (2021) revolves on the same idea, but using a combination of a CNN and a LSTM to predict not only the SYM-H index but also the complementary ASY-H index. With their LSTM + CNN model, they forecast both indices 1- and 2-hr ahead. With respect to our work, their architecture is different in that we use a standard LSTM model. Recent work by Iong et al. (2022) studies the forecasting of the SYM-H index 1- and 2-hr ahead by using a ML model belonging to ensemble methods, called gradient boosting machines (GBMs) (i.e., using an open-source regularizing gradient boosting framework; the eXtreme Gradient Boosting (XGBoost) library, Chen & Guestrin, 2016), obtaining very good performance as well. With respect to our work, their architecture is also different. Consequently, we take as our reference works along this paper the studies of Siciliano et al. (2021), Collado-Villaverde et al. (2021) and Iong et al. (2022). When possible our results are compared with these three works.

Other works also predict either just the SYM-H index (Cai et al., 2010) or the SYM-H and ASY-H indices about 1-hr ahead (Bhaskar & Vichare, 2019) using a non-linear autoregressive exogenous (NARX) model (Leontaritis & Billings, 1985). The latter work, as the one developed by Collado-Villaverde et al. (2021), concludes that the ASY-H index is significantly harder to forecast than the SYM-H index. On the other hand, Bailey et al. (2022) aim at forecasting the geoelectric field with LSTMs as well. While Pinto et al. (2022) forecast the ground magnetic field time derivative with LSTMs, Madsen et al. (2022) forecast both the ground magnetic field and its time derivative with LSTM networks and hybrid LSTM + CNN.

Concerning the hyper-parameter optimization, from all the works mentioned above the only one we were able to find where a thorough optimization was done is the work by Iong et al. (2022). They use the Nevergrad toolbox Rapin and Teytaud (2018) while we use the Optuna framework (Akiba et al., 2019) (see Section 4.1). In both cases, we use a gradient-free optimization approach, which is commonly used for hyper-parameter optimization for ML models where little or no information is known about the structure of the function to optimize.

Regarding the estimation of uncertainties in the prediction coming from a deep-learning model, several authors have pointed the importance of considering uncertainties to provide an optimal solution in the context of SW. The work developed by Gruet et al. (2018) uses an LSTM model combined with a Gaussian process (GP) model (Rasmussen & Williams, 2005) to provide up to 6-hr ahead probabilistic forecasts of the Dst index. This work points out the importance of probabilistic forecast against single-point predictions, concluding that their hybrid methodology provides improvements in the multi-hour ahead forecast of geomagnetic storms. Another work by Chakraborty and Morley (2020) considers a LSTM with deep GP regression to generate 3-hr ahead predictions of global geomagnetic index Kp with an associated probability distribution, also demonstrating that this architecture can be used to build probabilistic SW forecast models with good performance. In other context, Siddique et al. (2022) provides a survey of methods for uncertainty quantification using ML models, considering two example cases on how ML methods can quantify uncertainties within the domain of space physics. In particular, they consider the horizontal component of the perturbed magnetic field measured at the Earth's surface for the study of GICs, by training a NN with Bayesian inference (simply known as Bayesian NN) to generate predictions, along with epistemic and aleatoric uncertainties. The authors highlight the importance of the uncertainty evaluation to quantify a certain level of reliability on the model's existing accuracy. Licata et al. (2022) use a regularization

Table 1
List of the Geomagnetic Storms Split in Sub-Data Sets Used for Training, Validation and Test

Label	Start date	Duration (days)	SYM-H (nT)
Training sub-data set			
TR1	14/02/1998	8	−119*
TR2	02/08/1998	6	−168*
TR3	19/09/1998	10	−213
TR4	16/02/1999	8	−127*
TR5	15/10/1999	10	−218
TR6	09/07/2000	10	−335
TR7	06/08/2000	10	−235*
TR8	15/09/2000	10	−196*
TR9	01/11/2000	14	−174*
TR10	14/03/2001	10	−165*
TR11	06/04/2001	10	−275
TR12	17/10/2001	10	−210
TR13	31/10/2001	10	−313
TR14	17/05/2002	10	−113*
TR15	15/11/2003	10	−488
TR16	20/07/2004	10	−206
TR17	10/05/2005	10	−302*
TR18	09/04/2006	10	−110*
TR19	09/12/2006	10	−211*
TR20	01/03/2012	10	−149
Validation sub-data set			
V1	28/04/1998	10	−268
V2	19/09/1999	7	−160
V3	25/10/2003	9	−427*
V4	18/06/2015	10	−207*
V5	01/09/2017	10	−144*
Test sub-data set			
T1	22/06/1998	8	−120
T2	02/11/1998	10	−179*
T3	09/01/1999	9	−111
T4	13/04/1999	6	−122
T5	16/01/2000	10	−101*
T6	02/04/2000	10	−315
T7	19/05/2000	9	−159*
T8	26/03/2001	9	−434
T9	26/05/2003	11	−162*
T10	08/07/2003	10	−125*
T11	18/01/2004	9	−137*
T12	04/11/2004	10	−393*
T13	10/09/2012	25	−138
T14	28/05/2013	7	−134

technique to quantify the uncertainty in a thermospheric prediction model. In particular, the regularization technique they use is the dropout method (Gal & Ghahramani, 2016), which is one of the two methods explored in this work to estimate the uncertainties on the predicted SYM-H index (see Section 4.3).

From all these works, it is pretty clear that uncertainties play a fundamental role in prediction models in the context of SW and forecasting models of geomagnetic indices are not exceptions. In general, these works emphasize the relevance of providing associated uncertainties in order to provide reliable predictive models and thus understand the range of possible outcomes and to make more informed decisions. To the best of our knowledge, no other work has thoroughly evaluated yet the associated uncertainty in the prediction of the SYM-H index.

3. Data Set Selection and Processing

Following the same criteria as our reference works, the data set used in this corresponds to a sample of 42 geomagnetic storms that occurred between 1998 and 2018 (i.e., distributed along two solar cycles), see Table 1. These geomagnetic storms were recorded at ground-based geomagnetic observatories, and were preceded by changes in the magnetic field and plasma parameters of the interplanetary medium, which were also measured at the L1 Lagrange point by NASA's Advanced Composition Explorer (ACE) spacecraft. The intensity of the geomagnetic storms is defined by the SYM-H index which can be considered as a proxy of the response of the Earth's magnetosphere (especially the ring current) to solar activity and it is computed from data of a network of six magnetic observatories distributed in longitude across the low and middle-latitude region, with a time resolution of 1 min and precision of 1 nT. All the geomagnetic storms selected have a minimum SYM-H index lower than −100 nT, so they can be considered as either severe or extreme storms (Patowary et al., 2013). This ensures a high signal-to-noise ratio. Indeed, 55% of all these geomagnetic storms (23 out of 42) have a minimum SYM-H value between −200 and −100 nT, while the rest (i.e., 19 geomagnetic storms) have a minimum SYM-H value below −200 nT.

We follow the commonly used hold-out method for training ML models which is the process of dividing the full data set into three splits, one for training the model (48% of total sample) and other two splits to validate (12%) and test (40%) the trained model, respectively. The training sub-data set is used to train the LSTM model, the validation sub-data set stops the network training and prevent over-fitting, while the test (also known as hold-out) sub-data set is used as a proxy to evaluate the performance of the model on unseen data. This technique helps to identify weaknesses and guarantee the model to perform and generalize well on new and unseen data, even in the presence of various perturbations, such as data noise or changes in the distribution of the times-series data. The three sub-data sets are uniformly populated in terms of geomagnetic storm intensity and complexity, the latter measured by the presence of multiple depressions of the magnetic field. The length of the time interval of the considered geomagnetic storms range from 6 to 25 days, with an average of 10 days. This choice allows us to consider not only the main phase periods but also the initial and recovery phases, as well as previous and later quiet periods. This split, as shown in Table 1, was proposed by Siciliano et al. (2021) and followed by later works (Collado-Villaverde et al., 2021, 2023 (since the latter work has been published very recently, it has not been considered as reference work for comparison); Iong et al., 2022).

Table 1
Continued

Label	Start date	Duration (days)	SYM-H (nT)
T15	26/06/2013	8	−110
T16	11/03/2015	10	−233
T17	22/08/2018	12	−205

Note. The most relevant information is shown: label assigned to the storm, its starting date, its duration in days and its minimum value of the SYM-H index. Storms with multiple depressions are marked with the * symbol.

Following the same data set and split as previous works also allows an easy comparison of different results.

As already mentioned by Siciliano et al. (2021), a larger number of geomagnetic storms can be considered, as done for instance by Cai et al. (2010) or Bhaskar and Vichare (2019), though the additional ones are just either weak or moderate geomagnetic storms, adding no further predictive power to our LSTM model. This is due to the fact that all storm phases including quiet periods are already considered in all three sub-data sets.

The input variables (commonly named "features" in ML) used for training our LSTM model are the squared value of the magnitude of the IMF (B^2), the squared value of the y component of the IMF $\left(B_y^2\right)$ and the value of the y component of the IMF (B_z) (all these in Geocentric Solar Magnetospheric (GSM) coordinates recorded at L1 Lagrange point by the ACE satellite) and the SYM-H geomagnetic index, all expressed in nT. This selection of features are shown in Table 2 and are indeed the same as the ones used by Siciliano et al. (2021). However, the other two reference works do not use the same input features. On the one hand, Collado-Villaverde et al. (2021) use the magnitude and standard deviation of the IMF strength (B_t) and the magnitude and standard deviation of the x, y, and z GSM components of the IMF (B_x, B_y, B_z) together with SYM-H index values, all expressed in nT. On the other hand, Iong et al. (2022) consider several set of input features depending on their trained model. The minimal set uses B_x, B_y, B_z and SYM-H index values, all expressed in nT, while the maximal set uses the same as the minimal set together with solar wind plasma parameters (in particular, proton density (ρ) in cm^{-3}, the x component of the proton speed (V_x) in km s^{-1} and the proton temperature (T) in Kelvin) and derived quantities (in particular, the solar wind dynamic pressure and the electric field). The forecasted variable is the SYM-H index, same as in our three reference works.

All data are extracted from the NASA's OMNIWeb page (https://omniweb.gsfc.nasa.gov) with time resolution of 5 min (Papitashvili & King, 2023b). Although the data are available with a resolution of 1 min (Papitashvili & King, 2023a), the election of a lower resolution allows to reduce the computation time without a priori reducing predictive power, allowing also a direct comparison with the results of our reference works (since in these works they also use this time resolution). The 5 min sample is computed by averaging the 1 min samples, so that the data at minute 0 corresponds to the average from minutes 0 to 4.

The IMF variables are propagated to the nominal magnetospheric bow shock following the method described in the OMNIWeb site (King & Papitashvili, 2005). It is important to note in this context that in this study (and also in that of our reference works) the information available on SYM-H at the Earth's surface is assumed to be simultaneous with that of the IMF projected at the bow shock. However, the spacecraft measuring the IMF is located upstream of the solar wind at the L1 Lagrange point, which allows these data to be known some time in advance (typically between 15 and 60 min). This advantageous position, which is therefore not exploited here, is expected to have a significant role in the efficiency of the SYM-H predictions.

The IMF data overflows are removed from the full sample of geomagnetic storms and the remaining empty gaps are filled using a linear interpolation method. Geomagnetic storms generally have short periods with IMF data overflows but in few cases (e.g., training storm TR13 and validation storm V3) the overflows are large and occur near the peak of the storm activity. While linear interpolation is one possible way to address the problem of gaps, it is clearly not optimal when these are located in periods of high activity. We are aware that more sophisticated approaches could be exploited (e.g., the interpolation schemes proposed by Qin et al. (2007) or Marsal and Curto (2009) or even a ML-based method we are currently developing), but we used the same approach as our reference works to perform a direct comparison between their results and ours. It was decided to keep the storms with overflows near maximum of activity in the study, both to train and to stop training the network, since they represent a real possibility when part of the data is lost due to measurement errors or overflows or even detector failures.

Table 2
Input Features Used to Train Our Long Short-Term Memory Model

Training variables (nT)	B^2	B_y^2	B_z	SYM-H
Forecasted variable (nT)				SYM-H

Note. Same as in Siciliano et al. (2021).

Each geomagnetic storm time series is standardized using the associated mean and standard deviation, similarly to Siciliano et al. (2021) and

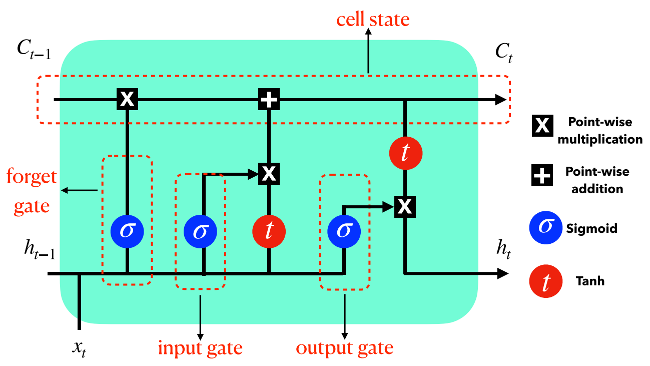

Figure 1. Long short-term memory (LSTM) "cell" based on four interacting layers (cell state, input gate, forget gate and output gate). An LSTM network consists of repetitions of such a cell for every step t of the time series.

Collado-Villaverde et al. (2021) (in the work developed by Iong et al. (2022), features in the input data do not need to be standardized before training when using GBMs as tree-based methods are invariant to monotonic transformations of the features so it is better equipped to handle inputs on different scales). However, in our case, the mean and standard deviation are just estimated from the training sub-data set, given that this is the only data previously known for the study, avoiding therefore the introduction of bias into the model. With the standardization the values are centered around the mean and given a unit standard deviation using the equation:

$$X_S = \frac{X - \mu_{TR}}{\sigma_{TR}}, \tag{1}$$

where X_S is the standardized time-series data, X is the original time-series data, μ_{TR} is the mean of the training series and σ_{TR} its standard deviation. The standardization of time-series data helps to improve the stability and convergence of the training as the model is less sensitive to the scale of features, learning the patterns in the data more effectively and without the necessity of spending much time adjusting to the different scales of the data. It also improves the performance because the model is less likely to over-fit the training data, as it is less sensitive to the specific values of the data.

4. Deep-Learning Model

Humans, as intelligent beings, do not have to learn how to speak, walk or cook from scratch every time. Our previous experiences on these tasks endure, and the ability to do it improves after several repetitions. Traditional NNs do not have this feature, and it is a major deficiency for some specific tasks such as time-series forecasting or natural language processing. Recurrent neural networks (RNNs) (Rumelhart & McClelland, 1987) address this problem. RNNs integrate cyclic connections, allowing information to persist, making them a more powerful tool to model sequential data than the traditional feed-forward NNs. LSTM networks are a variety of RNNs, capable of learning long-term dependencies and also forgetting irrelevant information, especially in sequence prediction problems. RNNs read the data sequentially, and attribute higher weights to the recent information. The back-propagation training of RNNs suffers from the so-called "vanishing gradient" problem, which limits the learning capabilities of the network. LSTMs address this problem by recognizing between long-term and short-term memory through a gating mechanism that regulates the flow of information. This allows the model to selectively retain or forget information based on its relevance, making it more robust and able to handle complex sequential patterns. In this work we use one such type of NN with a typical architecture shown in Figure 1, which we briefly describe below.

An LSTM network consists of a series of non-linear transformations for each time step with shared parameters (weights). The transformation for a generic input x_t at time step t is schematically represented in Figure 1. For each t, there are two outputs: the cell state c_t and the hidden output h_t, which are computed from four quantities. The first quantity is the result of the "forget gate" f_t, consisting typically of a sigmoid function applied to the linear function $W_f x_t + U_f h_{t-1} + b_f$. Here x_t is the input, h_{t-1} the hidden output of the previous time step, while W_f and U_f are the weight matrices and b_f is the bias vector. The motivation for the forget layer -whose value is a number in the range [0,1]—is to decide how much to "forget" about the previous time step's cell state c_{t-1} (i.e., "0" forgets the previous data and "1" uses the previous data). The second quantity is the "cell input" $\tilde{c}_t$, similar in structure to the forget gate, but with its own set of weights W_c, U_c, and b_c, and using a tanh function instead of a sigmoid. It represents the new information that would potentially be included in the new cell state c_t. How much of such information to be retained is determined by the third quantity: the result of the "input gate" i_t, defined as another sigmoid transformation, but with its corresponding weights W_i, U_i, and b_i. With the above three quantities and their interpretations, the new cell state is defined in an intuitive way as: $c_t = f_t \times c_{t-1} + i_t \times \tilde{c}_t$, with $c_0 = 0$ by definition. Finally, the fourth quantity is the result of the "output gate" o_t, being yet another sigmoid with weights W_o, U_o, and b_o. It has the role of a weight factor for the hidden output, computed as $h_t = o_t \times \tanh(c_t)$, with $h_0 = 0$.

The above series of transformations are consecutively applied to a finite number of time steps, from $t - l_b$ to t, with l_b being an optimizable hyper-parameter called the "look-back". The final aim is to predict an observable y_{t+l_f} at a future time $t + l_f$, with l_f being the "look-ahead," representing how much time in advance we want to make a prediction, which is fixed by the domain needs. In order to make such a prediction, right after the last LSTM cell applied on t and resulting in the hidden output h_t, a number of dense network layers are applied to transform the vector h_t into the predicted value of y_{t+l_f}, while potentially adding more expressiveness to the model. The number of dense layers is also another hyper-parameter to be optimized.

4.1. Hyper-Parameter Optimization Procedure

Hyper-parameter optimization is an essential ingredient when training state-of-the-art ML models, due to their high complexity. Traditional random- or grid-search strategies have shown to be very inefficient when the hyper-parameters space is large. Modern libraries exist which implement efficient algorithms for optimizing costly functions. One of the most popular ones, which we adopted here as already discussed in Section 2, is the Optuna framework. In particular, we employ a sequential model-based optimization algorithm that uses a tree-structured Parzen estimator (TPE) to approximate the objective function (Bergstra et al., 2011). Once the TPE model (probabilistic model that predicts the value of the objective function at a given set of hyper-parameters) of the objective function is built, it is used to select the next set of hyper-parameters to evaluate. We optimize the following hyper-parameters: the number of dense layers, the number of units of these layers, the learning rate, and the look-back parameter of the LSTM layer.

Finding multiple local minima can be a problem in hyper-parameter search. Because of the stochastic nature of gradient descent during training, there can be times when two identical trials result in a value of the loss function that varies more greatly than trials with different hyper-parameters would. For this reason, each trial is repeated five times. The mean and standard deviation of the loss function results for each trial with a set of hyper-parameters are calculated. Having done this, all trials with standard deviations that overlap with the best (i.e., lowest) mean of all trials are labeled as *best trials*. This procedure allows us to explore flat directions in the hyper-parameter space.

4.2. Robustness of the LSTM Model

Ensuring the robustness of state-of-the-art ML models, used to analyze and predict non-linear time-series data, is also of critical importance for reliable and effective decision-making, especially on those with direct impact on society. This requires careful design as well as rigorous testing and validation, but the benefits in terms of reliability and effectiveness are significant. Non-linear time-series data can exhibit complex and dynamic behavior, making it challenging to model and predict accurately.

As in many other works, we ensure the robustness of our LSTM model by using the hold-out validation technique and populating uniformly the three sub-data sets with storms with different intensity and complexity, as already discussed in Section 3. However, the performance estimate of the ML model may be highly dependent on the particular data set split used. If the split is not representative of the overall data set distribution, then the performance estimate may be biased. In our case, though no bias is expected, we further evaluate this potential issue by reshuffling the original list of geomagnetic storms in the three sub-data sets (see Table 1). Thus, we populated the new training sub-data set with the 17 storms from the original test sub-data set plus three storms from the original validation sub-data set. To populate the new test sub-data set, we used 17 storms from the original training sub-data set. Finally, the new validation sub-data set was filled with the remaining three storms from the original training sub-data set plus two validation storms from the original validation set. A visual representation of the baseline and reshuffled lists of geomagnetic storms used for training, validation and test is shown in Figure 2. With the new reshuffled list of the geomagnetic storms, we trained an alternative LSTM model (with its own optimized hyper-parameters) to forecast 1-hr ahead the SYM-H index and we obtained compatible performance without observing over-fitting, under-fitting or biases.

Other techniques could also be used to enhance the robustness of LSTM models such as more sophisticated flavors of cross-validation (e.g., k-fold cross-validation used in the context of SW by Zhelavskaya et al. (2019)), data augmentation, model ensembling, adversarial training or regularization. We will deeply explore this in future

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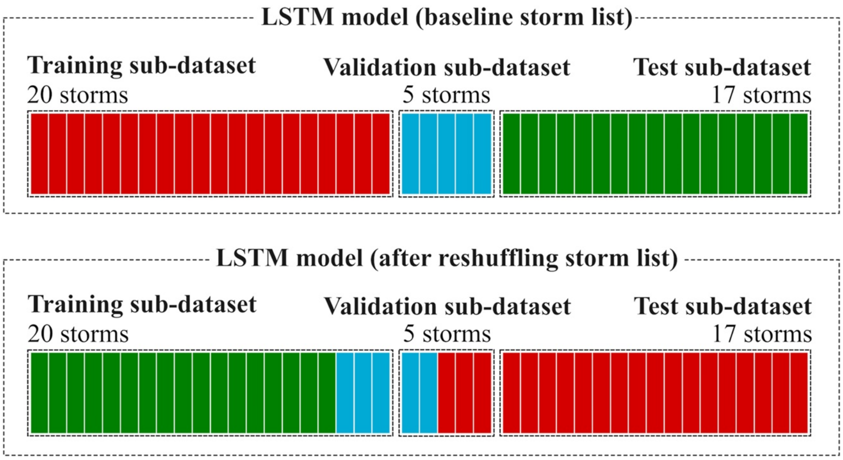

Figure 2. Visual representation of the two lists of geomagnetic storms used for training, validation and test for the baseline and the alternative long short-term memory models. The upper diagram represents the baseline list, as shown in Table 1, while the lower diagram represents an alternative ordering where the list of storms is reshuffled. Each rectangle represents a different storm. Colors are assigned based on the baseline list of the geomagnetic storms, where red, blue and green represent the original storms in the training, validation and test sub-data sets, respectively.

works. In any case, we want to point out that the regularization technique is indeed used in this work for the estimation of uncertainties (see Section 4.3 and Appendix A).

4.3. Estimation of Prediction Uncertainties

The quantification of uncertainties is a fundamental aspect of scientific research and data analysis, endorsing the reliability, precision, and interpretability of the obtained results, allowing therefore to make robust conclusions, develop accurate models, and advance in our understanding of nature. Therefore, we consider that forecasting models that provide not only predictions but also estimates of their associated uncertainties are indispensable to enable informed decision-making, risk mitigation, and the advancement of scientific understanding in the complex realm of SW.

To estimate the uncertainties associated with our predictions, two main approaches can be followed: a frequentist approach, in particular adopting the "bootstrapping" method, or a Bayesian approach, where several state-of-the-art methods can be adopted depending on the needs and scope such us the "dropout" method. In this work we use variations of the two mentioned methods, and compared the results between them.

4.3.1. Block-Bootstrapping Method

The bootstrapping is a method for estimating the statistical uncertainty by repeatedly re-sampling the "real" (observed or simulated) data set used for training, obtaining synthetic data sets out of this real data set, and recalculating the statistics (Efron, 1979). This is indeed equivalent to re-sampling from the empirical distribution (i.e., without making any assumptions about the underlying distribution of the real data set), instead of assuming a particular parametric shape of the likelihood.

The traditional bootstrapping generally fails with time series because the randomly re-sampling procedure breaks off the time dependence that concatenates adjacent samples in the data set. Thus, a special consideration has to be made for our case; if we can divide the data set in chunks of samples (i.e., blocks), and perform the block-bootstrapping re-sampling procedure on these blocks instead of on the individual samples, we can preserve the time dependence up to the division of the blocks. The ML community tends to use this method and additionally considering replacement ("with replacement" means that a particular block of the real data set can appear more than once in the synthetic data set) since this increases the versatility of the bootstrapping method even when the data is not normally distributed. Specifically, we use a block-bootstrapping procedure with replacement that works as follows: (a) divide the original time-series data (i.e., B^2, B_y^2, B_z and SYM-H) in the training sub-data

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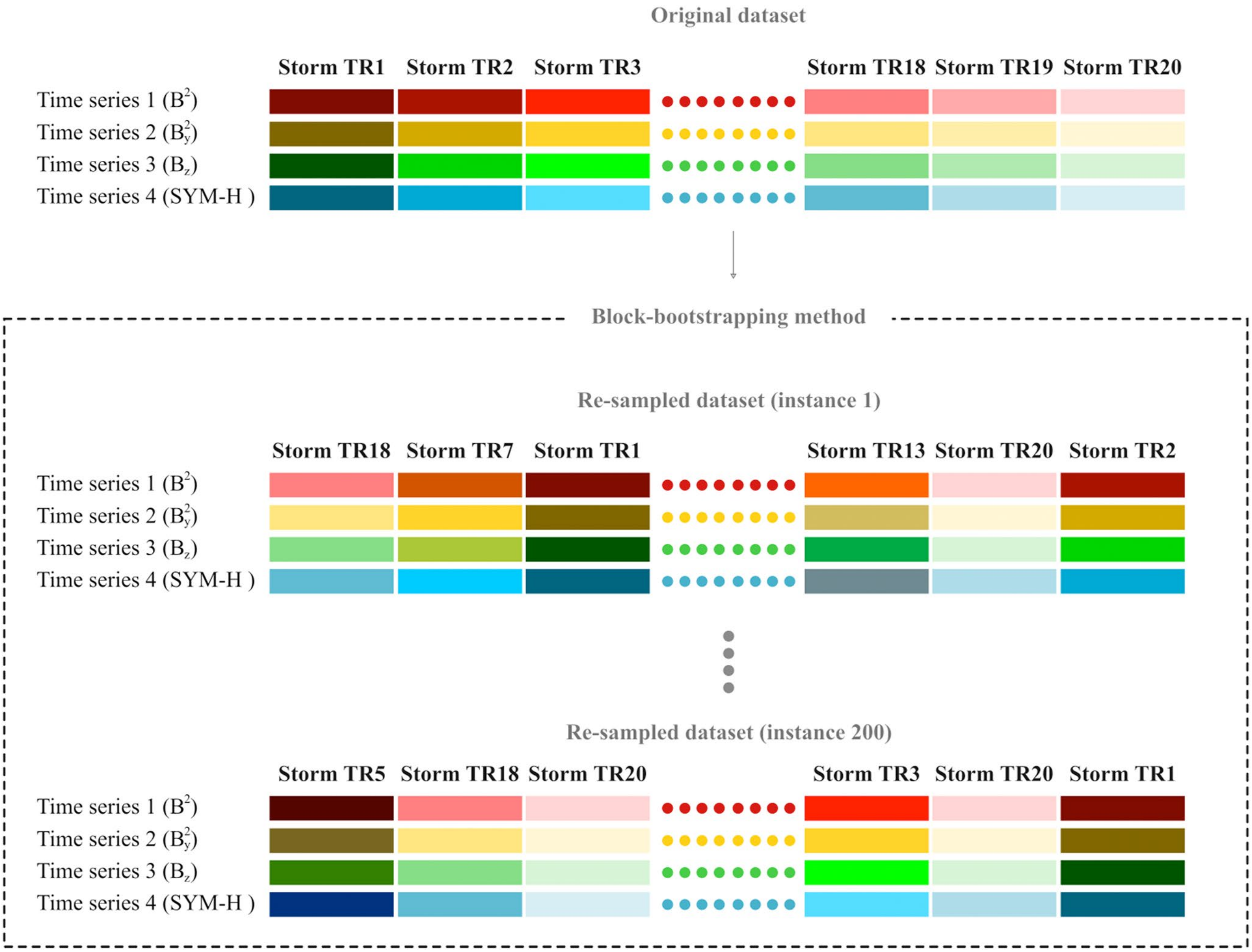

Figure 3. Schematic view depicting the block-bootstrapping with replacement procedure followed in this work. From the original time-series data, a re-sampling of the storms (represented by colored rectangles) is performed over the training sub-data set (i.e., over the 20 storms, shown in Table 2). This re-sampling is performed 200 instances and each re-sampled data set contains exactly 20 blocks (i.e., 20 storms). To illustrate that the same storm can appear more than once in a re-sampled data set due to the replacement procedure used, TR20 appears twice in the sketch for the instance number 200.

set into non-overlapping blocks (where block = geomagnetic storm), (b) generate a random sample of block indices with replacement from the set of all block indices. This means to re-sample with replacement 20 storms out of the 20 storms (i.e., each re-sampled sub-data set contains exactly 20 storms, where certain storms cannot appear and other storms can appear more than once in certain re-sampled sub-data sets). This forms a new set of block indices, which can be used to select the corresponding blocks from the original data, (c) concatenate the re-sampled blocks to form a new block-bootstrapped (i.e., re-sampled) data set and a train a new model, (d) repeat steps 2 and 3,200 instances to obtain 200 re-sampled data sets and therefore 200 trained models, (e) from these models we calculate the estimations (such as the mean or variance) of the uncertainties of prediction values. See Figure 3 for a schematic depiction of the procedure. Similar variations of traditional bootstrapping have already been used in the context of SW (e.g., Morley et al., 2018).

4.3.2. Concrete-Dropout Method

As discussed in Section 2, Bayesian approaches such as the GP model have been used in similar contexts (e.g., Chakraborty & Morley, 2020; Gruet et al., 2018), providing probabilistic predictions. In the case of deep NNs one of the most popular strategies is the so-called "dropout" method (Gal & Ghahramani, 2016). This regularization technique, that can be used to estimate model uncertainties, has already been exploited in the SW context (e.g., Licata et al., 2022). Roughly speaking, the idea consists in randomly turning off units of the different NN layers (this process is also referred to as Monte Carlo dropout). This has an immediate utility as a regularizer procedure; this is the reason for which dropout is commonly used at the training phase in order to control over-fitting. However, as pointed out by Gal and Ghahramani (2016), such a procedure is mathematically equivalent to a variational inference algorithm, with a specific choice of the variational distribution. In particular, if dropout is also used at the test phase, the probability distribution of the predictions would be equivalent to the ones that would be obtained by computing the standard predictive distribution of the Bayesian approach, under the chosen variational approximation.

In this work, a particular combination of the standard dropout method and a variation, known as concrete dropout (Gal et al., 2017), is used in the dense layers to estimate the prediction uncertainties. An essential parameter in the dropout implementation is the dropout probability p. Formally, p is the probability for a Bernoulli (binary) random variable to take value equal to 1; so by sampling from the Bernoulli distribution, once for every unit in a hidden layer, such a unit is turned off with a probability of $1 - p$. Traditionally, p is considered as an important hyper-parameter to be optimized, for example, by grid-search, which can be computationally expensive in largely parameterized models. This is the motivation behind concrete dropout, which modifies the traditional dropout algorithm in such a way that p becomes an optimizable parameter during the normal training period. This is done by modifying the loss function so that it has an explicit -and differentiable- dependence on p, which is the result of approximating the Bernoulli distribution by its continuous relaxation using the concrete distribution. In our NN architecture, we have implemented the concrete-dropout method for the dense layers following the LSTM layer, and consequently the associated dropout probability p is automatically optimized during the training process. However, for the LSTM layer itself we stick to the traditional implementation of dropout, where the parameter p is in this case included as an hyper-parameter optimizable with the Optuna procedure. The resulting optimal value for 1-hr ahead prediction for the LSTM dropout probability is $p = 0.0128$.

4.3.3. Some Considerations

We evaluate both the block-bootstrapping and the concrete-dropout methods to estimate the prediction uncertainties in our forecasting LSTM-based model and we show the comparison in Appendix A. We find that there are small difference in performance between both methods though the uncertainties on the critical periods of the storm regions (i.e., peaks) are usually larger for block bootstrapping, having a slightly better coverage. This behavior tends to be the opposite on calm regions (away from the peaks). Therefore considering that block bootstrapping slightly perform better in the peak regions, we have set block bootstrapping as the baseline procedure to quantify uncertainties in this work.

A note of caution is needed at this point. When reporting our prediction uncertainties, we are more specifically reporting the systematic (or epistemic) uncertainties of the expected (mean) values of the SYM-H index, which we calculate as the output of our LSTM network. Note that this is not the same as the total prediction uncertainties, which include the data noise (also known as aleatoric, or statistical uncertainties), and which we do not have available. Since the reported epistemic uncertainties decrease as the number of data points increase, it is perfectly consistent to have a very precise determination of the mean predictions of the SYM-H index, describing data whose noise is appreciably larger (and consequently having values beyond the corresponding interval of epistemic uncertainties). On the other hand, we are also not considering in this work the uncertainties related to the other input variables (related to the IMF B and its components), that can have non-negligible contributions as the work developed by Morley et al. (2018) concludes. This work studies and quantifies the effect of uncertainties in solar wind input measurements driving on the predictions of the SYM-H index and ground magnetic disturbances for a given geomagnetic storm, demonstrating their importance for SW applications. While we plan to include them in a future work, we nonetheless expect their impact on our results to be small (compared to the model uncertainties), after checking that the uncertainties in B^2 are of few percent.

5. Results and Discussion

After discussing the data and analysis setup in previous sections, we turn now to present the results of our analysis. In Section 5.1 we present the optimization of hyper-parameters needed to learn the evolution of the SYM-H index, the relative importance of each parameter in the final result and explore the possible correlation between hyper-parameters. The input feature importance is discussed in Section 5.2. Then, we present the overall perfor-

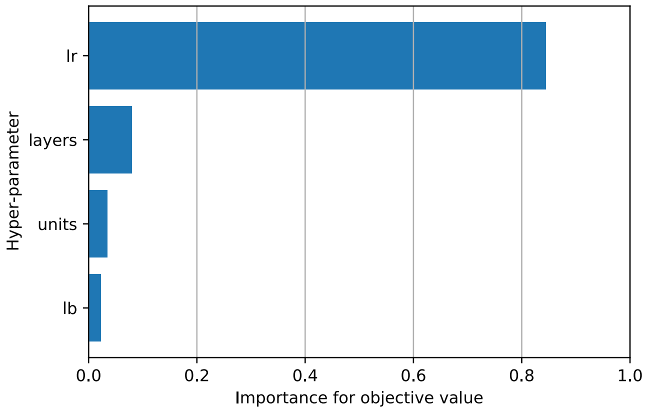

Figure 4. Hyper-parameter importance bars for 1-hr ahead prediction for learning rate (`lr`), number of dense hidden layers (`layers`), number of units in all hidden layers (`units`) and look-back (`lb`).

mance of the trained algorithm when predicting the evolution of the SYM-H index multiple-hour ahead (in particular, for six values: 0.5-, 1-, 1.5-, 2-, 3.5-, and 5-hr ahead) in Sections 5.3 and 5.4. Additionally, a comparison with our reference works for 1- and 2-hr ahead predictions is also shown and discussed. Finally, in Section 5.5 the effects of some systematics uncertainties are evaluated and discussed.

Similarly to our reference works, to evaluate the accuracy of the predictions of the LSTM for forecasting the SYM-H index, the root mean squared error (RMSE) and Coefficient of Determination R^2 defined by Equations 4 and 3, are used.

$$\mathrm{RMSE}(y,\hat{y}) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i-\hat{y}_i)^2}, \tag{2}$$

$$R^2(y,\hat{y}) = 1 - \frac{\sum_{i=1}^{N}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{N}(y_i-\bar{y}_i)^2}, \tag{3}$$

where N is the sample size, y_i the observed value, $\bar{y}_i$ the mean value of the evaluated sample and $\hat{y}_i$ the predicted value. The RMSE metric provides insight into how well the prediction matches the observation on average. The lower the value, the better the performance of the model is. While R^2 considers the amount of variance of the observed data, explained by the prediction. A value of 1 means that the model perfectly predicts the observed data. On the other hand, a 0 value implies that the model is equivalent to forecasting the mean of the observed variable. R^2 could reach negative values if the predicted model performs worse than simply predicting the average of the observed data. These metrics are commonly used to calculate the goodness of prediction models (Liemohn et al., 2018), therefore we use them to compare models.

5.1. Hyper-Parameter Optimization

Once we have chosen an LSTM as the basic architecture for the time-evolution analysis, the next step is to optimize the hyper-parameters of the learning structure. We use the approach discussed in Section 4.1. This optimization is done independently for all six `x`-hour ahead predictions shown in this work. The results of the hyper-parameter optimization for 1-hr ahead prediction are shown in Figures 4 and 5.

In particular, we vary the number of fully-connected layers which are placed after the LSTM architecture (`layers`), the number of neurons of these dense hidden layers (`units`) and the learning rate parameter (`lr`). We also explore different values of the look-back parameter (`lb`), the amount of previous data we allow the network to explore in order to predict the future evolution; this value is reported in terms of number of 5 min steps, unless otherwise indicated.

The ranges in which each hyper-parameter is optimized are summarized in Table 3. The ranges in the second column stand for the 1-hr ahead prediction while the ranges for the other multiple-hour ahead predictions are shown in the fourth one. We also provide the values chosen after the optimization process for 1- and 2-hr ahead predictions. Variations of each of these parameters are not equally important, as shown in Figure 4, for 1-hr ahead prediction. Indeed, we found that the learning rate (`lr`) is key in out trained models, whereas variations of the depth and width of the fully-connected layer (`n_layers`, `n_unit`) are much less important. This indicates that, once the LSTM is learning the time series, the particular characteristics of the additional dense layer are not that relevant. We also found that, in the range we explored, the look-back parameter was not an important handle. This would indicate that we have already chosen an optimal look-back range. Note, however, that if we were interested in describing other, less global, parameters than the SYM-H index, the look-back parameter may change. This optimization is only valid for the output prediction we have chosen to describe.

The best hyper-parameter values (in terms of mean-square error (MSE)) for 1-hr ahead prediction according to Optuna are: (`n_layers, n_unit, lr, lb`) = (4, 386, 3.12×10^{-5}, 75 steps), which are shown in Table 3.

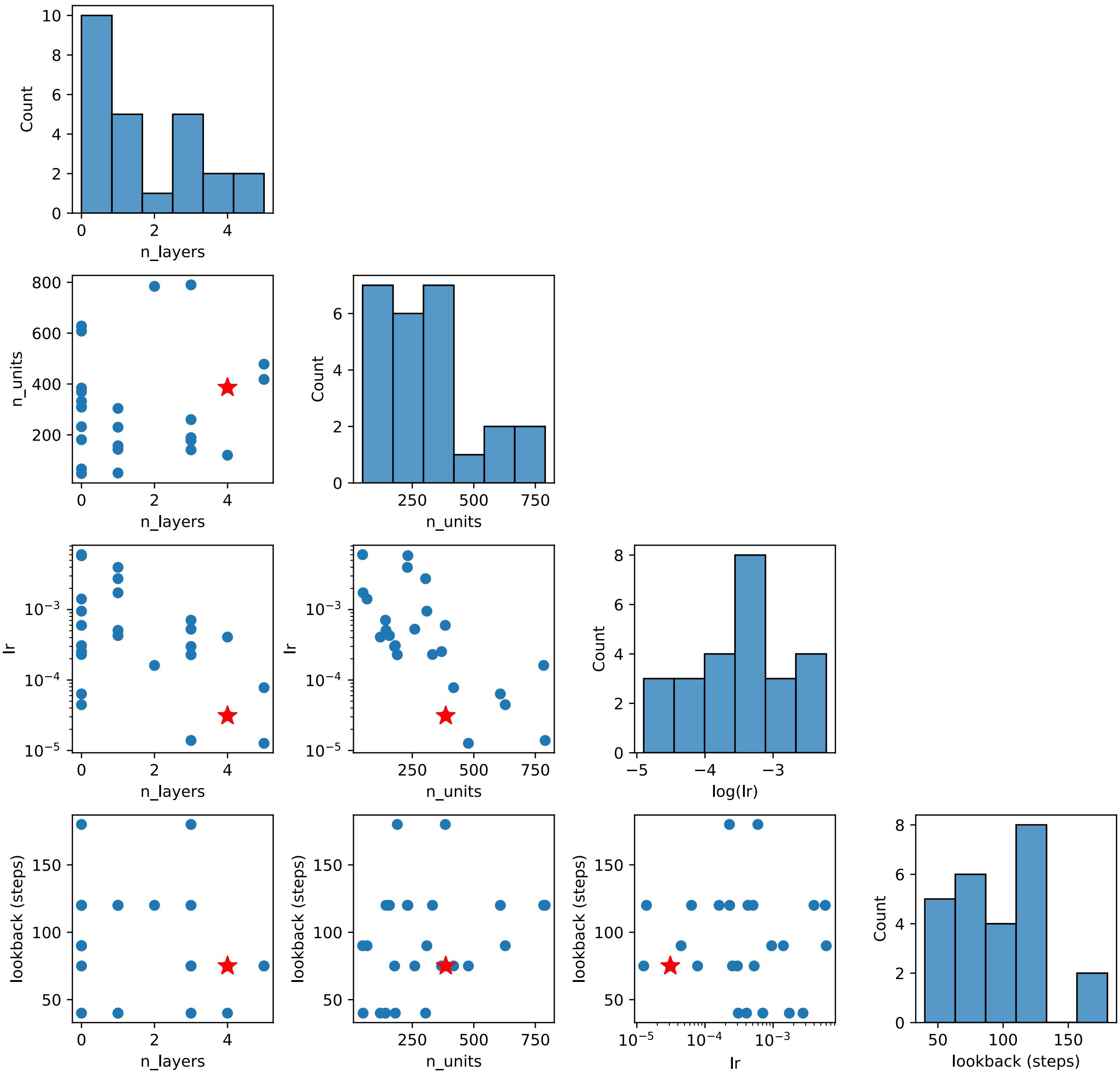

Figure 5. Pair-wise scatter-plots for the hyper-parameters optimized via Optuna for the long short-term memory architecture for 1-hr ahead prediction. Out of the total of 25 cases, the blue points correspond to hyper-parameter values which cover the minimum value of the mean-square error that results from using the global optimum values. Red stars indicate the reported optimum values. The histograms along the diagonal, for each hyper-parameter, are the result of marginalizing all the points from the rest of hyper-parameters.

Table 3
Ranges in Which Each Hyper-Parameter are Optimized for Both the 1-hr Ahead and the Multiple-Hour Ahead Predictions, and Chosen Values for Both 1- and 2-hr Ahead Predictions

Hyper-parameter	Search range (1-hr ahead)	Chosen value (1-hr ahead)	Search range (other pred.)	Chosen value (2-hr ahead)
Number of layers	[0, 10]	4	[0, 20]	19
Number of units	[0, 1,000]	386	[0, 1,000]	57
Learning rate	$[10^{-6}, 10^{-1}]$	3.12×10^{-5}	$[10^{-6}, 10^{0}]$	2.44×10^{-5}
Look-back (steps)	[40, 75, 90, 120, 180, 360]	75	[40, 75, 90, 105, 120, 135, 150, 165, 180, 195, 210]	195

Table 4
Mean Root Mean Squared Error (RMSE) Values for 1-hr Ahead Predictions of the SYM-H Index for This Work With 95% CL Uncertainties, for Each of the Storms in the Test Sub-Data Set for Our Long Short-Term Memory (LSTM) Model, and Using Few Interplanetary Magnetic Field Variables and SYM-H Values as Input Features for the Training

	RMSE (nT)			
Label	This work LSTM	Siliciano et al. LSTM	Collado et al. LSTM + CNN	Iong et al. GBM
T1	**6.30 ± 0.70**	6.70	6.63	5.86
T2	**10.00 ± 5.00**	8.90	8.91	7.73
T3	**4.02 ± 0.40**	5.30	5.86	4.28
T4	**8.00 ± 4.00**	7.20	6.68	5.83
T5	**5.30 ± 0.90**	5.60	5.20	4.93
T6	**8.40 ± 1.70**	10.70	8.58	8.28
T7	**7.70 ± 1.20**	8.30	7.26	6.84
T8	22.00 ± 6.00	16.30	**13.34**	14.49
T9	**9.70 ± 0.50**	11.30	10.03	10.19
T10	**6.90 ± 0.40**	8.50	7.69	7.15
T11	**8.90 ± 0.60**	8.70	9.52	8.51
T12	19.00 ± 4.00	17.50	15.18	**14.55**
T13	**4.10 ± 0.40**	4.20	4.08	3.89
T14	**5.10 ± 0.60**	5.60	6.43	5.90
T15	**4.90 ± 0.50**	5.50	4.67	4.98
T16	9.40 ± 1.30	9.00	7.88	**7.56**
T17	5.89 ± 0.60	5.90	5.67	**5.03**
Total data set	9.57 ± 1.40	9.00	–	–
Mean	8.57	8.55	7.86	7.41
Std. dev.	1.19	0.90	0.71	0.76
p-value	–	6.10^{-6}	4.10^{-4}	0.02

Note. The results of this work are compared with reference works, which use different deep-learning models. Bold represents best values as defined in the text. The *p*-value is also provided.

However, while these specific values are indicated as optimal, one should keep in mind that slightly different values could lead to the same performance; there could be "flat directions," that is, combinations of hyper-parameter values away from the reported optimum which produce equally low MSE. Most importantly the optimization made using Optuna assumes that the hyper-parameters are uncorrelated; indeed, the hyper-parameters may be correlated to some extent, while the procedure assumes complete independence.

We have also explored the impact of these caveats by performing a multidimensional scan of the hyper-parameters instead of assuming total uncorrelation. The results for 1-hr ahead prediction are summarized in Figure 5, where we show the pair-plots between the different hyper-parameters. The points shown in the scatter plots correspond to hyper-parameter values for which, upon repeating the trials five times, within their standard deviation, cover the minimum value of the MSE that results from using the global optimum values specified above. The optimum values are also shown. The histograms along the diagonal, for each hyper-parameter, are the result of marginalizing all the points from the rest of hyper-parameters. For these scatter plots we observe no evident correlation between pairs of hyper-parameters, which validates the use of the Optuna procedure.

However, the flat directions are explicitly present in almost all axes. For example, if using six hidden layers instead of one, while fixing the rest of hyper-parameters to values different from their "optimum," we get equally good results, statistically speaking. Analogously, this happens with the number of units per hidden layer, which can be as high as 800 (with respect to the reported optimum at 386), or the look-back parameter at 300 (with respect to the optimum at 75). In all cases we observe that each hyper-parameter can admit large excursions in combination of specific values of other hyper-parameters without sacrificing the figure of merit. This is nothing but the consequence of a highly complex parametric dependence of the loss function with respect to the hyper-parameters of the model, as is often the case with the large models used by the community nowadays. These conclusions apply to the hyper-parameter optimization of all six x-hour ahead predictions shown in this work.

5.2. Feature Importance

Neural networks are often considered black-box algorithms though some external inference techniques can be used to extract useful information that can help to understand deep-learning models. Computing feature importance in LSTM models is indeed an important aspect of model interpretation and understanding. Feature importance is a measure of how much a particular input variable (or feature) contributes to the output of the model. Indeed, understanding feature importance can help to identify and select the input features that are most relevant for a given prediction model. It can also provide valuable insights into the underlying patterns, dynamics and relationships present in the considered time-series data. There are several techniques that are commonly used to compute feature importance in LSTM models. Some of these techniques are the "input permutation" (Breiman, 2001; Fisher et al., 2019), "SHapley Additive exPlanations" (Lundberg & Lee, 2017), "Leave-One-Feature-Out," "gradient-based method," "layer-wise relevance propagation," and "activation-based methods" among others.

In this work, the approach used to compute the feature importance in our LSTM model is based on the "input permutation" technique. We repeated the training procedure, using the same optimized hyper-parameters already discussed in Section 5.1, but adding disturbances in the input data (i.e., in IMF data and SYM-H values). Thus, for each of the four input features, the values of all of the other features were shuffled, new predictions were calculated using the original test sub-data set, and RMSE was calculated. This procedure was performed 15 times for each variable; this is a total of 60 training model instances. The average value of the RMSE for each case is compared to a baseline value calculated with no shuffling (i.e., with the average RMSE value of the RMSE values shown in Table 4). The output of the feature importance results for 1-hr ahead prediction is shown in Figure 6.

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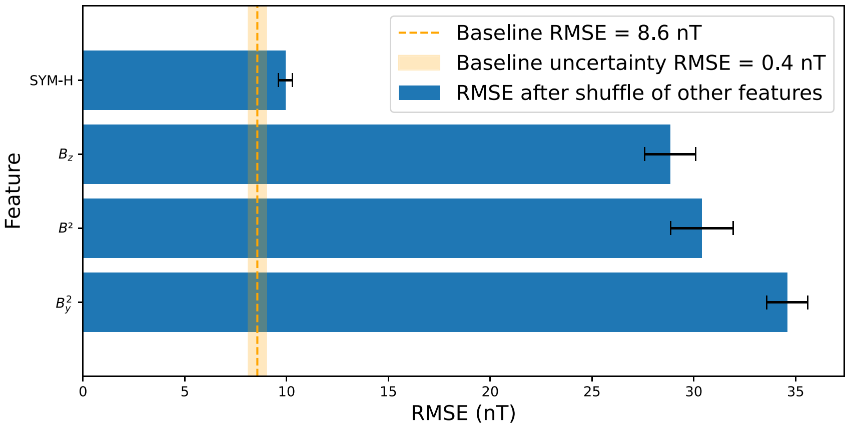

Figure 6. Ranking of the feature importance for 1-hr ahead prediction using an approach based on the "input permutation" technique (the smaller the value, the more important the variable is). Each bar represents the mean value of the root mean squared error evaluated over all test storms after having shuffled all except the indicated feature variable. The uncertainty bars represent the standard deviation, and the vertical orange line represents the baseline value calculated with no shuffling along with its own standard deviation (that can be computed by averaging values from Table 4).

In the followed method, the most important features are the ones that, when all other variables are shuffled, result in an RMSE closer to the baseline average RMSE value. Thus, from the obtained results, we conclude that SYM-H values represent the most important feature for our LSTM model, similarly to other works (e.g., Siciliano et al., 2021).

It is important to point out that the interpretation of feature importance in LSTM models can be challenging, as these models are inherently complex and exhibit dynamic and non-linear behavior. Additionally, the results can be influenced by the data pre-processing (e.g., interpolation approach for data gap filling), the choice of input scaling and normalization as discussed in Section 3, as well as the choice of model architecture and the optimization of the training hyper-parameters as discussed in Section 4.1.

5.3. 1- and 2-hr Ahead Predictions of the SYM-H Index

In this section, we present the 1- and 2-hr ahead predictions of the SYM-H index. These predictions are obtained by using our LSTM model, with optimized hyper-parameter, over the test sub-data set (i.e., over 17 test storms), using the metrics discussed above. Associated uncertainties are also provided, allowing us the validation of the forecasting model and the understanding of its reliability. These uncertainties are estimated using the block-bootstrapping method, discussed in Section 4.3.1, by producing 200 model instances to produce 200 samples of predictions and obtain estimations of the uncertainties of prediction values. For all these instances the RMSE and R^2 values are calculated for each geomagnetic storm. Then, from the samples of RMSE and R^2 values for each storm, a mean and a 95 confidence level (CL) are obtained (corresponding to 1.96σ).

Since we use the same data set (42 storms) and same training-validation-test splitting as in our reference works, a comparison is also performed against these works. As already discussed, we use a standard LSTM model as Siciliano et al. (2021) which allows a directly comparison for 1-hr ahead predictions. For the other two reference works, despite the differences highlighted along this paper, we also include their predictions in our 1- and 2-hr ahead prediction comparisons.

Tables 4 and 5 show the values of RMSE (in nT) and R^2, respectively, for the 1-hr head predicted SYM-H index and their uncertainties at 95% confident level (CL) for each storm of the test sub-data set. The RMSE and R^2 obtained by forecasting the full test sub-data set and mean RMSE (considered each storm as an independent data set) are also reported. Figure 7 shows the same 1-hr head results but graphically, where the blue dots represent the average of our predictions and the blue symmetric segments correspond to the 95% CL of the predictions for both RMSE and R^2. The reference works are also shown in this figure. The R^2 values for Iong et al. (2022) are not shown as these values are not report by these authors.

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Table 5
Mean R^2 Values for 1-hr Ahead Predictions of the SYM-H Index for This Work With 95% CL Uncertainties (Estimated by Using the Block-Bootstrap Method), for Each of the Storms in the Test Sub-Data Set for Our long Short-Term Memory (LSTM) Model, and Using Few Interplanetary Magnetic Field Variables and SYM-H Values as Input Features for the Training

	R^2		
Label	This work LSTM	Siliciano et al. LSTM	Collado et al. LSTM + CNN
T1	**0.870 ± 0.050**	0.890	0.870
T2	**0.920 ± 0.070**	0.940	0.939
T3	**0.969 ± 0.008**	0.950	0.936
T4	**0.910 ± 0.080**	0.930	0.922
T5	**0.951 ± 0.013**	0.950	0.946
T6	**0.969 ± 0.010**	0.960	0.971
T7	**0.944 ± 0.020**	0.950	0.953
T8	**0.910 ± 0.060**	0.960	0.965
T9	**0.810 ± 0.025**	0.750	0.798
T10	**0.925 ± 0.008**	0.900	0.907
T11	**0.887 ± 0.014**	0.890	0.864
T12	**0.946 ± 0.031**	0.960	0.966
T13	**0.941 ± 0.011**	0.940	0.939
T14	**0.959 ± 0.008**	0.960	0.932
T15	**0.964 ± 0.006**	0.950	0.966
T16	0.954 ± 0.013	0.960	**0.969**
T17	**0.966 ± 0.009**	0.970	0.968
Total data set	0.944 ± 0.020	0.960	–
Mean	0.929	0.930	0.930
Std. dev.	0.010	0.013	0.011
p-value	–	1.10^{-4}	5.10^{-5}

Note. The results of this work are compared with other reference work, which use different deep-learning models. Bold represents best values as defined in the text. The *p*-value is also provided.

By quantifying uncertainties in our forecasting model, we can assess the validity of our model by comparing our results to reference predictions. Anyhow, there are some caveats as uncertainties are not available for the predictions provided by the reference works. First, Siciliano et al. (2021) results correspond to the best forecast among 20 single-point predictions per test storm with random initialization weights, and not such selection is done in our work (as we provide uncertainty estimation). Similar approach is presumably followed by Collado-Villaverde et al. (2021) and Iong et al. (2022). With the purpose of comparing, we can define our best value at 95% CL as the lower (upper) bounds of the RMSE (R^2) prediction interval. One can appreciate that our results are generally compatible with these of our reference works at 95% CL. For the RMSE metrics our best values are slightly better for the full test sub-data set and for 10 out of 17 storms our mean RMSE values improve their best values. Similar results are obtained for R^2 metric.

Comparing with Siciliano et al. (2021) results, we can conclude that our architecture leads to slight better performance within 95% CL using the same LSTM model, which we believe is mainly a manifestation of the achieved optimization of the hyper-parameters. Comparing with Collado-Villaverde et al. (2021), only two storms (T8 and T16) improve our best RMSE results and only storm T16 improves our best R^2 result. Finally, results from Iong et al. (2022) show better results in terms of RMSE for storms T8, T12, T16, and T17 if we compare against our best values. From these results based on the selected metrics, we conclude that using additional input features to the ones we already use and/or adopting other deep-learning models do not help to improve forecasting performance.

In addition, 2-hr ahead predictions are also compared to two of our reference works. Table 6 shows our results for the RMSE and R^2 metrics for 2-hr ahead predictions at 95% CL together with single-point prediction results by Collado-Villaverde et al. (2021) (for RMSE and R^2) and Iong et al. (2022) (for RMSE only). The RMSE and R^2 obtained by forecasting the full test sub-data set and mean RMSE (considered each storm as an independent data set) are also reported. Figure 8 shows graphically the same 2-hr head results. We see that for 7 of the 17 test storms our best values are slightly better than those of Collado-Villaverde et al. (2021) and 6 out of 17 when comparing with Iong et al. (2022). In both cases their RMSE values tend to fall consistently near or at the lower side of our CL interval. It becomes clear that either the LSTM + CNN and GBM architectures or considering additional input features or their combination is slightly more appropriated for forecasts beyond 1 hr.

It is worth stressing that the aim in this work is not to build and optimize a robust model in terms of prediction performance, but to study the effect of a hyper-parameters optimization and the importance of prediction uncertainties to achieve a highly reliable forecasting model. Nonetheless, as we see, we still can give very competitive results for 1-hr (and even 2-hr) ahead predictions. We highlight the importance of considering model uncertainties when making predictions with statistical significance and comparisons with other results. The uncertainty can vary greatly from storm to storm, to the point of that predictions could became irrelevant if uncertainties exceed certain values. Considering this point is crucial if one finally wants to generate an alarm system, make reliability assessments, and take decisions about potential risks at critical infrastructures.

Finally, a paired *t*-test is performed to check if the differences among mean RMSE values across the test sub-data set are statistically significant at 95% CL. Accordingly the *p*-values reported in Tables 4–6, we may conclude that our model is statistically compatible with all others for 1-hr ahead prediction but not with any other model for 2-hr ahead. Same result is reported by Iong et al. (2022) when they compared with the other two reference works. Anyhow, for further and much precise comparison it would be essential the evaluation of uncertainties

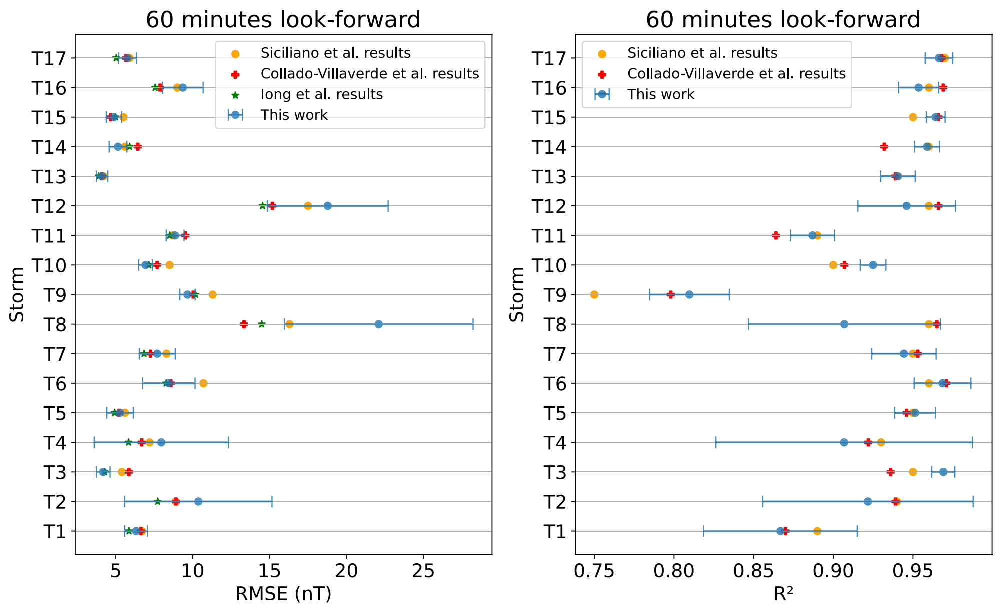

Figure 7. Root mean squared error (RMSE) and R^2 values for the 1-hr predicted SYM-H index for each one of the 17 test storms. The results of this work are shown in blue with 95% CL uncertainty bars (estimated by using the block-bootstrap method), while those from Siciliano et al. (2021), Collado-Villaverde et al. (2021), and Iong et al. (2022) are shown in orange circles, red crosses and green stars, respectively, which correspond to their best RMSE and R^2 results.

for all reference models (which are not available for the time being), to perform a more appropriated and robust statistics test, for instance a χ^2 test.

5.4. Multiple-Hour Ahead Predictions of the SYM-H Index

In order to further study the evolution of the prediction uncertainties with the look-forward parameter, multiple-hour ahead prediction models are optimized and built. The quantitative results in Figures 7 and 8 are rather global measures of performance, as they evaluate the goodness of predictions during the whole storm. Moreover, we are indeed more interested to know how well the forecasting model is performing during shorter periods of time, primarily during the peaks of activity. To illustrate this point, in Figures 9 and 10, we show the observed and multiple-hour ahead predictions with their associated uncertainties of the target variable SYM-H index (in nT) for test storms T11 and T12. As shown in Table 1, T12 is one of the most extreme storms in the test sub-data set with a SYM-H of −393 nT, while T11 has very moderate activity (with a SYM-H index of −137 nT), both with multiple peaks of activity. The multiple-hour ahead predictions for all 17 geomagnetic storms in the test sub-data set can be found in Figures S36 to S52 in Supporting Information S1.

In these figures, the orange band represents the 95% CL of the predictions coming from the block-bootstrapping procedure, while the mean prediction is shown by the red dashed lines, and the actual test values are shown as a solid blue line. For each storm we also show the residuals (labeled as "Data-Model"), which are computed just subtracting the prediction mean from the observed values, and orange bands representing the Gaussian-distributed uncertainty from the 200 model instances we performed using the block-bootstrap method. Each SYM-H distribution with its bottom panel shows a different multiple-hour ahead prediction (i.e., considering a look-forward parameter value of 0.5, 1, 1.5, 2, 3.5, and 5 hr). We observe a very good agreement between the predictions and the observations for 30 min and 1 hr (even for 90-min ahead predictions). As expected the largest residuals are around peaks of highest activity. Note how the prediction uncertainties are also larger around the peaks, as one could anticipate. For larger look-forward parameter values, both residuals and the uncertainties increase remarkably, more drastically for the most intense storm (i.e., T12). Moreover, a displacement of the prediction with respect to the test data can be observed, increasing with the look-forward value, which obviously increases both the residuals and uncertainties. In fact, generally less intense storms tend to have better RMSE values, as seen

Table 6
Mean Root Mean Squared Error (RMSE) and R^2 Values for 2-hr Ahead Predictions of the SYM-H Index for This Work With 95% CL Uncertainties (Estimated by Using the Block-Bootstrap Method), for Each of the Storms in the Test Sub-Data Set for Our Long Short-Term Memory (LSTM) Model, and Using Few Interplanetary Magnetic Field Variables and SYM-H Values as Input Features for the Training

	RMSE (nT)			R^2	
Label	This work LSTM	Collado et al. LSTM + CNN	Iong et al. GBM	This work LSTM	Collado et al. LSTM + CNN
T1	10.7 ± 2.50	8.99	**8.28**	**0.600 ± 0.270**	0.766
T2	17.00 ± 5.00	13.42	**11.58**	**0.800 ± 0.090**	0.862
T3	7.50 ± 0.70	5.88	**5.65**	0.907 ± 0.015	**0.936**
T4	13.00 ± 4.00	9.31	**8.83**	**0.770 ± 0.090**	0.848
T5	**8.20 ± 1.60**	7.29	7.28	**0.890 ± 0.040**	0.894
T6	**13.80 ± 2.50**	12.44	12.61	**0.910 ± 0.050**	0.939
T7	11.60 ± 1.50	**8.94**	9.93	0.880 ± 0.040	**0.929**
T8	29.00 ± 7.00	**18.48**	24.52	**0.800 ± 0.150**	0.932
T9	**14.40 ± 0.80**	13.94	13.74	**0.610 ± 0.050**	0.612
T10	10.30 ± 0.80	9.93	**9.50**	**0.830 ± 0.030**	0.845
T11	13.00 ± 1.20	**12.06**	12.07	**0.750 ± 0.040**	0.782
T12	28.00 ± 7.00	**21.08**	22.23	**0.870 ± 0.110**	0.934
T13	6.60 ± 0.50	5.21	**5.15**	0.861 ± 0.022	**0.900**
T14	8.30 ± 0.70	**6.80**	7.39	0.900 ± 0.015	**0.924**
T15	6.00 ± 0.50	**5.28**	5.63	**0.945 ± 0.013**	0.957
T16	13.80 ± 2.90	**11.70**	12.12	**0.900 ± 0.040**	0.931
T17	9.60 ± 0.80	8.27	**7.97**	0.907 ± 0.021	**0.932**
Total data set	14.00 ± 1.80	–	–	0.874 ± 0.050	–
Mean	12.99	10.53	10.86	0.831	0.878
Std. dev.	1.70	1.08	1.31	0.025	0.022
p-value	–	0.43	0.17	–	0.26

Note. The results of this work are compared with two of our reference works, that is, from Collado-Villaverde et al. (2021) using an LSTM + CNN and from Iong et al. (2022) using a GBM. Bold represents best values as defined in the text. The *p*-value is also provided.

e.g., in the results of in Table 4). Also geomagnetic storms with large periods of calm, as T13 (see Figure S47 in Supporting Information S1), tend to have lower RMSE values since generally the prediction is more accurate outside the peaks within the uncertainty.

Finally, we quantify the evolution of the mean RMSE and its uncertainty as a function of the look-forward parameter. Figure 11 shows the mean RMSE (with 95% CL uncertainty band) for several look-forward values growing from 30 min to 5 hr for storm T11 (moderated storm), T12 (severe storm) and for the forecasting of the full data set. The RMSE grows almost linearly for 30-min to 2-hr look-forward values. For larger look-forward values the trend changes slightly and the uncertainties grow notably, making our predictions less meaningful or even useless. Indeed, the evolution of the model uncertainties shows that our predictions beyond 2-hr ahead are not reliable or robust enough to base an alarm system on. In addition, when forecasting 2 hr or more, the residuals are dominated by the time lag between the prediction and the test data, even for the calm periods. This is clearly shown in Figures 9 and 10.

5.5. Effects of the Systematics Uncertainties

To evaluate systematics uncertainties in the prediction, we explore their effects first on the residuals of the predictions and then on the coverage probability of the prediction values with respect to the observed values.

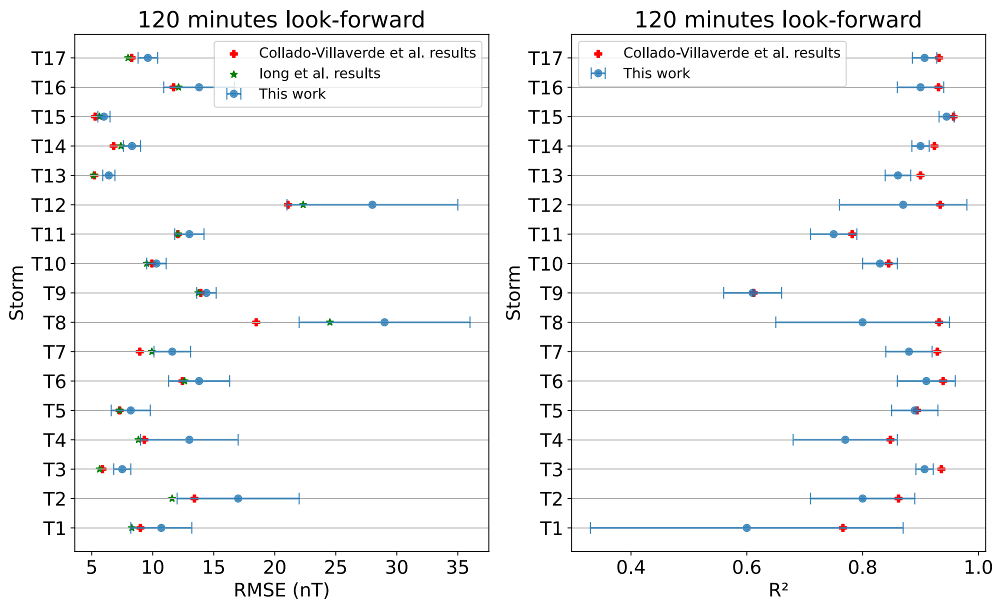

Figure 8. Root mean squared error (RMSE) and R^2 values for the 2-hr predicted SYM-H index for each one of the 17 test storms. The results of this work are shown in blue with 95% CL uncertainty bars (estimated by using the block-bootstrap method), while those from Collado-Villaverde et al. (2021) and Iong et al. (2022) are shown in red crosses and green stars, respectively, which correspond to their best RMSE and R^2 results.

We expect that our uncertainty estimation is underestimated, where the two suspects of missing uncertainty are the intrinsic fluctuations of SYM-H and the inherent *timing* problem in time-series predictions. The latter source of uncertainty can be identified by the lag of the predicted data with respect to the observed data. Indeed, this time lag can be seen in the literature forecasting geomagnetic indices. This is a common occurrence for LSTM models and other ML models handling sequential data (e.g., Gruet et al., 2018): usually it is a consequence of using a limited training sub-data set (in terms of storms or/and input features) for predicting time-series data with a significant auto-correlation, which can make sometimes difficult for the model to accurately identify the underlying patterns and trends.

The residuals for each test storm are calculated, considering the full storm, the interval peak (defined in Figures S18–S34 in Supporting Information S1), and out of the interval peak. The results are reported in Table 7 and are visualized in Figure S35 in Supporting Information S1. The data identified as out of the peak are intervals that correspond to periods of relative calm. Although we are aware that some activity occurs before a storm starts and also that some storms include multiple peaks. In general multiple peaks are included in the so called peak interval (a exception is storm T8 with large activity far from the main peak).

From the residuals in the out of peaks intervals in Table 7, we observe first that they are centered around 0.14 ± 0.42 nT, well below 1 nT (notice that 1 nT is the precision of the SYM-H data). More importantly, we note that the mean of the standard deviation for the residual of all storms in this interval is 4.22 ± 1.43 nT. This value can be interpreted as an intrinsic variation of periods of relative calm, and can aid us in estimating a correction to our original uncertainty values.

In the peak interval, the long tailed distributions are centered at 0.75 nT, with a higher dispersion (2.12 nT) than in other intervals. In these distributions is where we identify the effect of the systematics bias: the larger deviations around the peaks are mainly due to a difference in *timing* of the predictions with respect to the observations that vary from storm to storm (for more detail see Figures S18–S34 for individual storms and Figure S35 for a full picture of the training data set in Supporting Information S1). This effect is evident in the peaks of the storms where large negative (positive) residuals occur when there is a lag on the prediction as SYM-H index rapidly drops (recovers). This is most dramatic at the peaks of maximum storm activity. This is indeed the case at least for storms T1, T2, T4, T6, T8, and T17 featuring residuals around the peak in the range 20–100 nT for 1-hr ahead prediction.

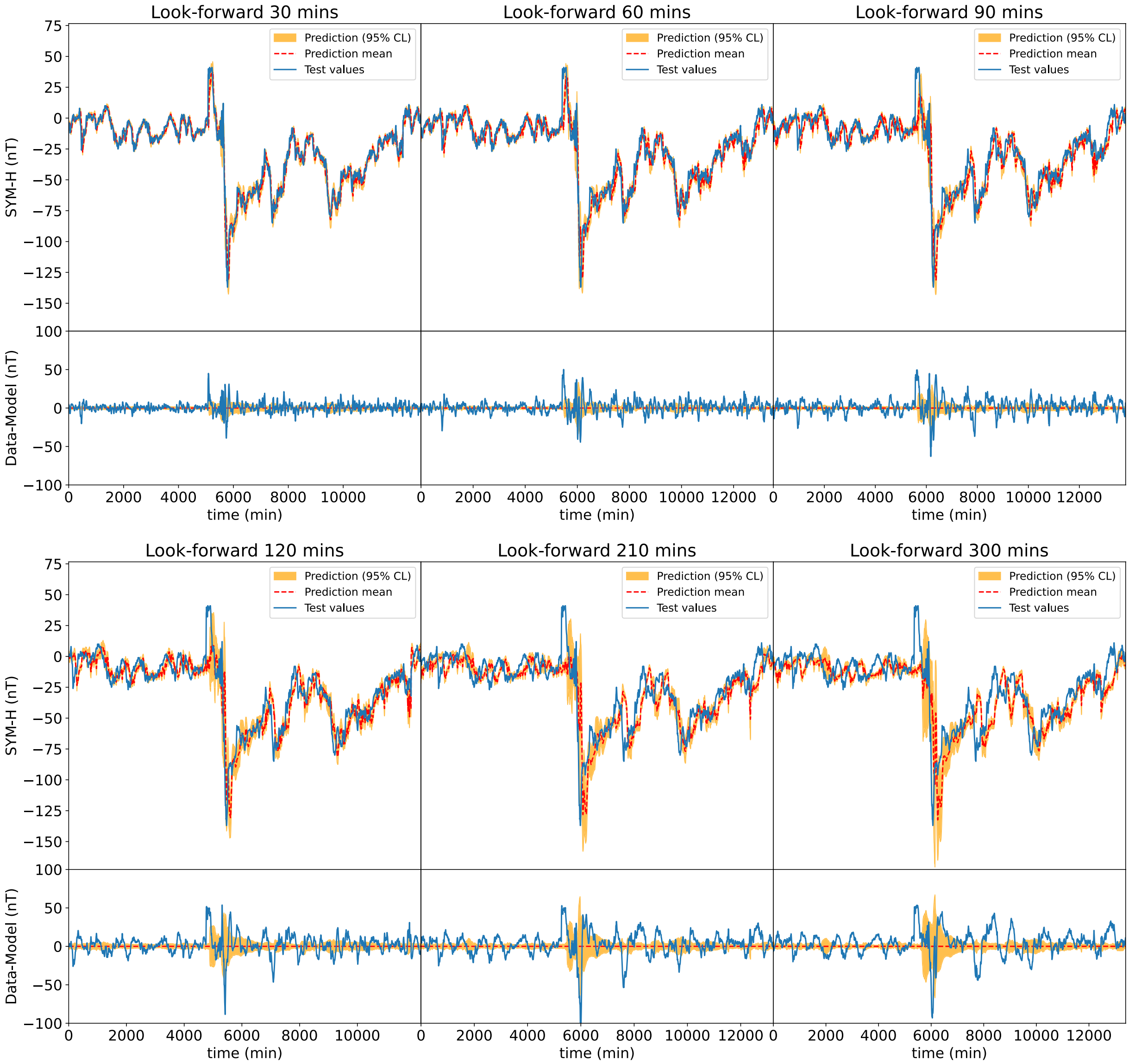

Figure 9. Observed and multiple-hour ahead predicted time-series distributions for the SYM-H index for the geomagnetic storm T11 of the test sub-data set. In all distributions, we show in a solid blue line the observed data (i.e., test data), in red dashed line the mean for the 0.5-, 1-, 1.5-, 2-, 3.5-, and 5-hr ahead predictions of the SYM-H index from our long short-term memory model, and in an orange band the 95% CL estimated by using the block-bootstrap method. The lower panels represent the residuals with respect to the model prediction mean.

To estimate quantitatively the effect of the systematic uncertainty identified in the residuals, we first calculate the coverage given by our current uncertainty estimation. To evaluate this value we use the prediction interval coverage probability (PICP) (Mistryukova et al., 2023). The PICP metric is defined as:

$$\mathrm{PICP} = \frac{1}{N}\sum_{i=1}^{N} m_i, \quad \text{with} \;\; m_i = \begin{cases} 1, & \mathrm{PL}_i^L \le t_i \le \mathrm{PL}_i^U, \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where N is sample size, PL_i^L and PL_i^U are the lower and upper limits on a predictor i and t_i is the i-element of the data. PICP is an estimation of the probability that a measure value lies within the prediction confident interval. In Figure 12 the PICP values for 1-hr ahead prediction are presented for all storm in test sub-data set (top), storm

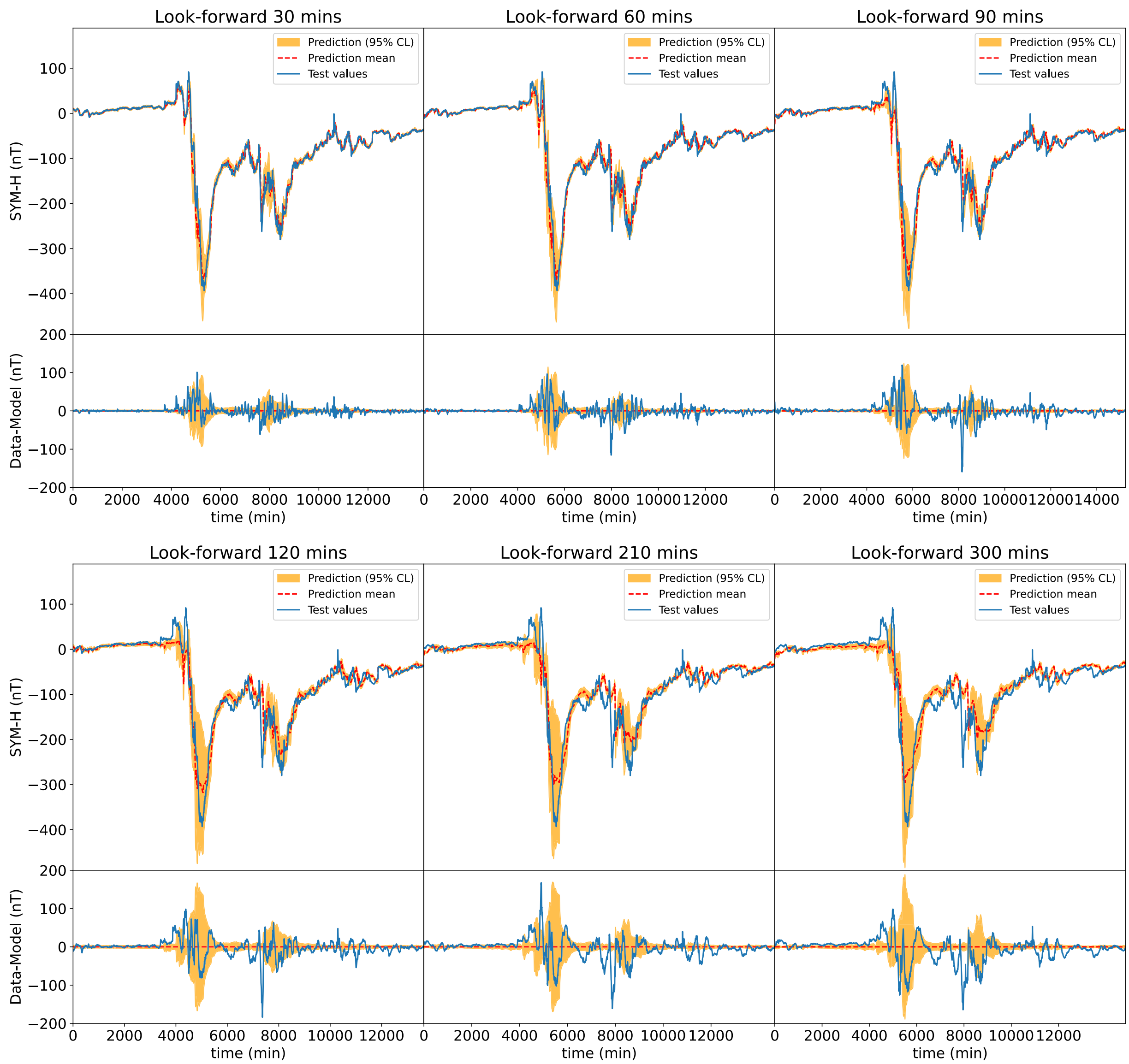

Figure 10. Observed and multiple-hour ahead predicted time-series distributions for the SYM-H index for the geomagnetic storm T12 of the test sub-data set. In all distributions, we show in a solid blue line the observed data (i.e., test data), in red dashed line the mean for the 0.5-, 1-, 1.5-, 2-, 3.5-, and 5-hr ahead predictions of the SYM-H index from our long short-term memory model, and in an orange band the 95% CL estimated by using the block-bootstrap method. The lower panels represent the residuals with respect to the model prediction mean.

T11 (middle), and T12 (bottom) for different uncertainty estimations and for given multiples of the standard deviations of the prediction values, considering both the full storm and the peak intervals.

So far in this work, we have defined the lower and upper limits of the uncertainty of the prediction as the multiple of the standard deviation corresponding to 95% CL. With these intervals, we expect to include close to 95% as PICP of the observed data. An ideal PICP value is the expected inclusion of values corresponding to each multiple of the standard deviation; for example, we expect a value of PICP of 68.3% CL (95.5% CL) when the width of the uncertainty is 1σ (2σ). This is represented by the dashed black line in the plots. When compared to this ideal PICP, we see that our uncertainty estimations (red dots in Figure 12) are underestimated. This result is not surprising. As commented in Section 4.3, the orange bands only represent the epistemic uncertainties (the uncertainties on the expected mean from the prediction model), reason for which there may be observed values lying outside the bands. This leads us to consider other sources of uncertainties.

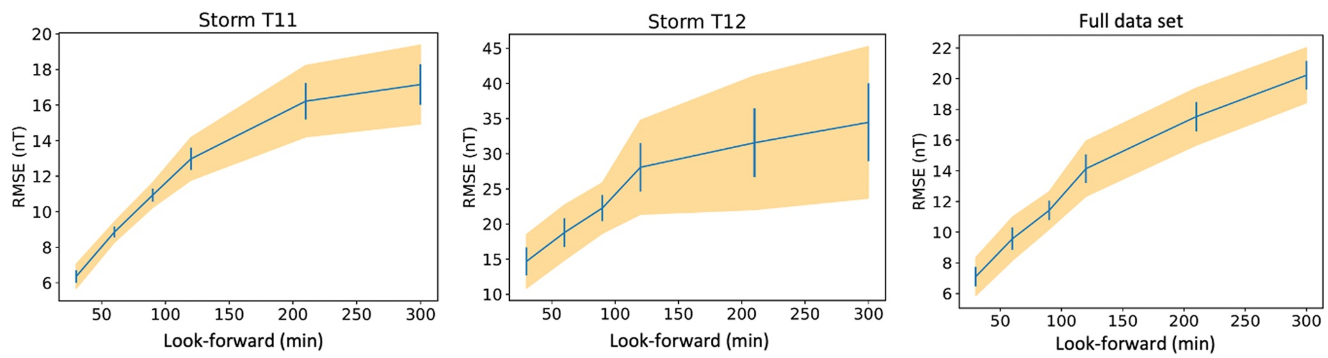

Figure 11. Mean root mean squared error with 95% CL uncertainties (orangish band), estimated by using the block-bootstrap method, as a function of the ahead time prediction for storm T11 (left), T12 (center), and full data set (right).

Other source of uncertainties could be the intrinsic variation of the SYM-H index, considered as independent from the uncertainties of the model (so far, uncertainties on the input features have not been considered). Recalling that we have retrieved an estimation of this value from the evaluation of the residuals in the out of peak interval, we can recalculate PICP after adding this value as a source of uncertainty. The bands in Figure 12 correspond to the PICP calculated after including a 4.22 ± 1.43 nT uncertainty on the predicted SYM-H data. For the full test sub-data set and full storm range, the PICP band is compatible with the perfect coverage, while for storms T11 and T12, or when only the peak interval are considered, *there is still a remaining underestimation*.

Table 7
Residuals for SYM-H Calculated as the Difference Between the Test Data ("Data") and the Mean of the 200 Block-Bootstrapping Instances ("Model"), for the Peak Region, Out of the Peak Region and for the Full Storm

	Residuals (data-model) (nT)					
	Peak interval		Out of peak interval		Full storm interval	
Storm	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
T1	0.83	8.47	0.60	4.76	0.66	6.03
T2	0.67	11.87	0.46	3.32	0.59	9.43
T3	0.43	5.39	0.08	3.19	0.19	7.20
T4	6.48	14.80	−0.35	3.19	0.89	7.42
T5	3.52	6.41	0.77	3.79	1.51	4.79
T6	−3.70	12.54	−0.11	4.65	−1.13	7.92
T7	0.18	11.95	−0.15	3.98	−0.06	7.12
T8	−1.89	38.03	0.58	8.33	−0.06	20.66
T9	0.42	11.85	0.53	6.19	0.48	9.39
T10	1.17	7.59	0.34	3.53	0.92	6.66
T11	2.08	10.27	0.48	5.53	1.34	8.45
T12	−0.19	22.58	−0.45	3.63	−0.32	16.19
T13	0.03	5.66	−0.18	3.31	−0.14	3.92
T14	1.37	7.68	−0.10	3.57	0.23	4.86
T15	0.18	7.25	−0.04	2.95	0.02	4.68
T16	0.50	18.02	0.47	5.09	0.47	8.71
T17	0.59	11.09	−0.59	3.08	−0.38	5.51
Mean	0.75	12.44	0.14	4.22	0.31	7.89
Std. dev.	2.12	8.00	0.42	1.43	0.66	4.40

As already mentioned, time shift is a common behavior of ML models handling time-series data. This peculiarity is of course another relevant source of uncertainty. To quantify this effect on PICP at 95% CL, we shifted the observed time-series with respect to the predicted values (1-hr look-forward) by different time values between 0 and 2 hr and calculate the hypothetical new PICP for each time shift. We have observed an improvement in the coverage metrics when the time shift is in the interval between 0 and 1 hr. This shows that the inherent time lag of the model may account for the underestimation of the uncertainty observed in Figure 12 because, after taking into consideration both sources of uncertainty, an ideal PICP coverage is obtained for various storms in the test sub-data set. A correction of this uncertainty is outside the scope of this work, given that a larger data set (more storms for training), more input features and maybe a more appropriate architecture should significantly mitigate this behavior. The recent work by Collado-Villaverde et al. (2023) is going in that direction including Attention to the deep-learning model and also other features with promising results in their of global metrics.

6. Conclusions

In this paper we present an LSTM model with associated uncertainties to predict the evolution of a geomagnetic index with a good level of reliability during a set of geomagnetic storms. We introduce a powerful method to optimize the hyper-parameters of the forecasting model. We also propose two statistically robust procedures, block bootstrapping and concrete dropout, to estimate the uncertainties of these predictions. In particular, we forecast the SYM-H index, a quantity whose variation during a storm is a good summary of its strength. As input features, we used IMF data (B^2, B_y^2 and B_z) from the ACE spacecraft located at the L1 Lagrange point together with SYM-H values.

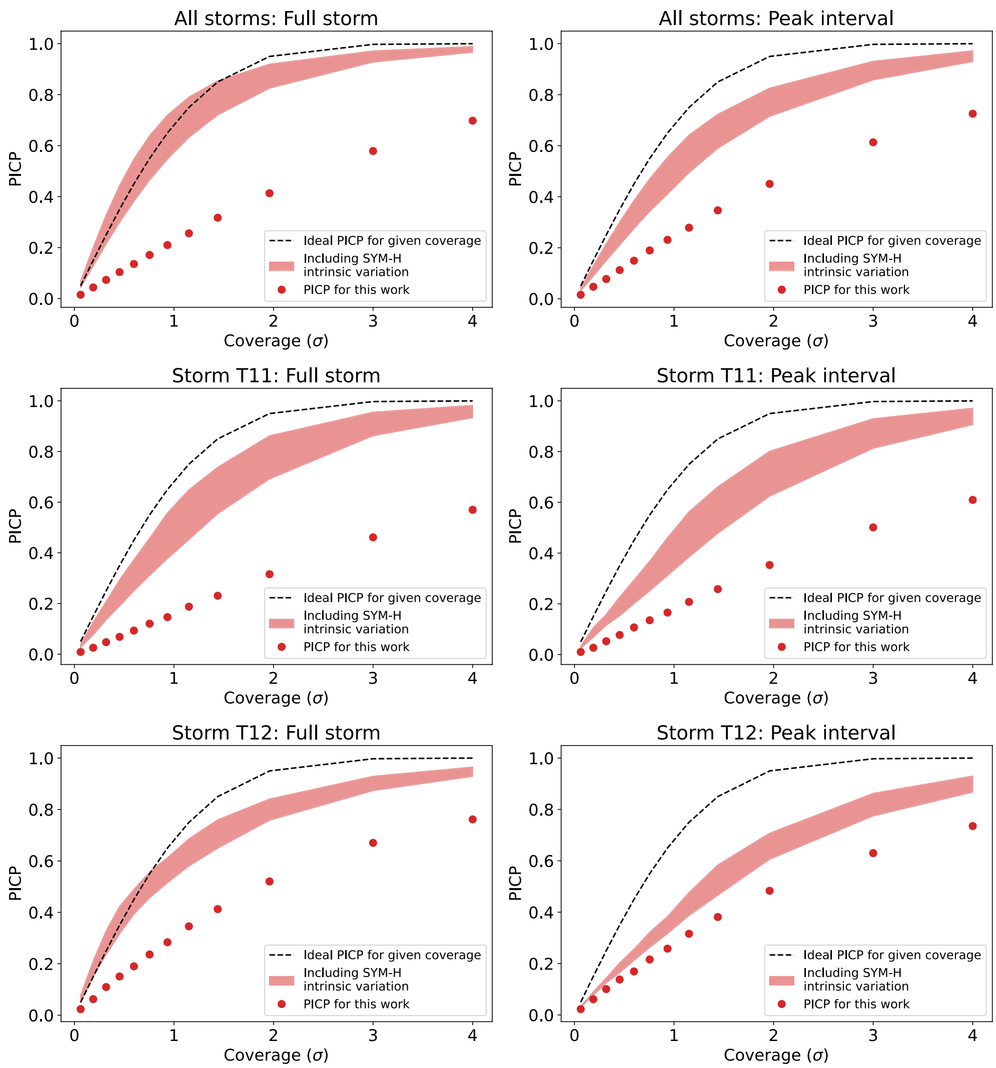

Figure 12. Evaluation of coverage using prediction interval coverage probability (PICP) metric for 1-hr ahead prediction for different standard deviations. Dots are the PICP values considering uncertainties of our model, the PICP values in the bands consider also the intrinsic variations of SYM-H within 4.22 ± 1.43 nT, and dashed line is the ideal PICP corresponding to the coverage. The first row represents the average of all storms, whereas the second and third rows represent the PICP values for storms T11 and T12, respectively.

We present the 1- and 2-hr ahead predictions of the SYM-H index with associated uncertainties estimated using the block-bootstrapping method, discussed in Section 4.3.1, by training 200 model instances to produce 200 samples of predictions and obtain estimations of the uncertainties of prediction values. We also compare the our best values, as defined in Section 5.3, for the RSME and R^2 metrics with the best single-point values from three reference works (Collado-Villaverde et al., 2021; Iong et al., 2022; Siciliano et al., 2021) that use different architectures (LSTM, LSTM + CNN, and GBMs). From a quantitative comparison on 1-hr ahead predictions, we conclude that our results are on average compatible with the results from our reference works and generally have slight better performance within 95% CL. We also see that the hyper-parameters optimization improves the performance. Furthermore, we also conclude that using additional input features to the ones we already use and/or adopting other deep-learning models do not help to improve forecasting performance. From comparisons on 2-hr ahead predictions, the tend is that results from the reference works using additional input features and/or alternative ML architectures (LSTM + CNN or GBM) are slightly more appropriated for forecasts beyond 1 hr.

However, we also study the behavior, goodness and evolution of the prediction uncertainties by means of the residuals (Data-Model) and the mean RMSE and its uncertainty, both as a function of the look-forward parameter, obtaining multiple-hour ahead prediction models. We show how the uncertainties grow remarkably with the look-forward values, specially around the peaks and also more drastically for the most intense storms, as expected. We conclude that for look-forward values larger than 2-hr ahead the uncertainties grow so much that our predictions are not reliable or robust enough to base an alarm system on, and take decisions about potential risks at critical infrastructures.

Finally, we have evaluated and quantify our uncertainty estimation using the residuals and the coverage probability. From these studies, we conclude that the estimated uncertainties are indeed underestimated since we are just considering the uncertainties of the model (epistemic) and other sources of uncertainty should be included, as intrinsic fluctuations of SYM-H, timing behavior of the time series, or uncertainties on the input features (out of the scope of this work).

Future work will explore the identified improvements mentioned along this paper, such as adding promising input features, use more sophisticated interpolation method to fill data gaps, improve the training sub-data set by exploiting cross-validation techniques, add an Attention layer in combination with our LSTM model and/or consider uncertainties associated to the input features among others.

Appendix A: Bootstrap and Dropout Methods for Estimating the Prediction Uncertainties

Two methods are used to estimate prediction uncertainties, block bootstrapping (a frequentist approach) and concrete dropout (a Bayesian approach). The description of the methods can be found in Section 4.3. To illustrate the features of those methods graphically, geomagnetic storms T7 and T8 from the test sub-data set are considered. As shown in Table 1, T7 has a moderate activity (with a SYM-H index of −159 nT), while T8 is the most extreme of the storms in the test sub-data set with a SYM-H of −434 nT. Additionally, the former has multiple peaks. Figure A1 shows for T7 and T8 the 1-hr ahead prediction with the associated uncertainty for SYM-H index, for the two methods, block bootstrapping (right) and concrete dropout (left). The display of all the 17 geomagnetic storms in the test sub-data set can be found in Figures S1–S17 in Supporting Information S1. For computation of the two approaches we use the residual of the prediction (blue line at the bottom of the plots) and the uncertainty (orange bands).

The first thing to notice is that both methods give similar residuals on average. In fact, the mean of the residuals, considering the full test sub-data set is −0.09 nT (with a standard deviation of 7.89 nT) when using block bootstrapping and 0.27 nT (with a standard deviation 8.08 nT) with concrete dropout, being therefore fully compatible. Moreover, the same level of compatibility between methods are observed for each individual storm.

However, when only the peak region of individual storms are considered, one of the methods, generally the block-bootstrap method, is slightly better than the other. Figure A2 shows a zoom-in around the peaks of maximum activity for the mentioned storms, T7 and T8 (all the 17 geomagnetic storms in the test sub-data set can be found in Figures S18–S34 in Supporting Information S1). Although this feature is not very obvious due to the different scale of the storms, the means of the residuals for storm T7 are similar for both methods (−2.11 nT for block bootstrapping and −1.51 nT for concrete dropout) while a large difference is obtain for T8 (−1.25 nT for block bootstrapping and −6.67 nT for concrete dropout).

Furthermore, concerning the prediction uncertainties (orange band in the figures), we see more differences. Typically the uncertainties on calm regions (away from the peaks) are usually larger for concrete dropout than for block bootstrapping. This trend is best seen in storm T7 (due to the scale). However, when considering the region around the peak the opposite trend is observed. As discussed in Section 5.5, these are only the uncertainties of the model ("epistemic") and other sources of uncertainty should be included, as intrinsic fluctuations of SYM-H, timing behavior of the time series, or any other source.

The critical periods of a storm are precisely when the peaks occur which provide in fact the more relevant information to derive an alarm. Also, considering that block bootstrapping slightly perform better in these regions, we have adopted it as the baseline procedure to quantify uncertainties in this work. This criteria is adopted also based on the slight better PICP coverage around the peak of the more severe storms. Although either of the methods could be used due to the small difference in performance.

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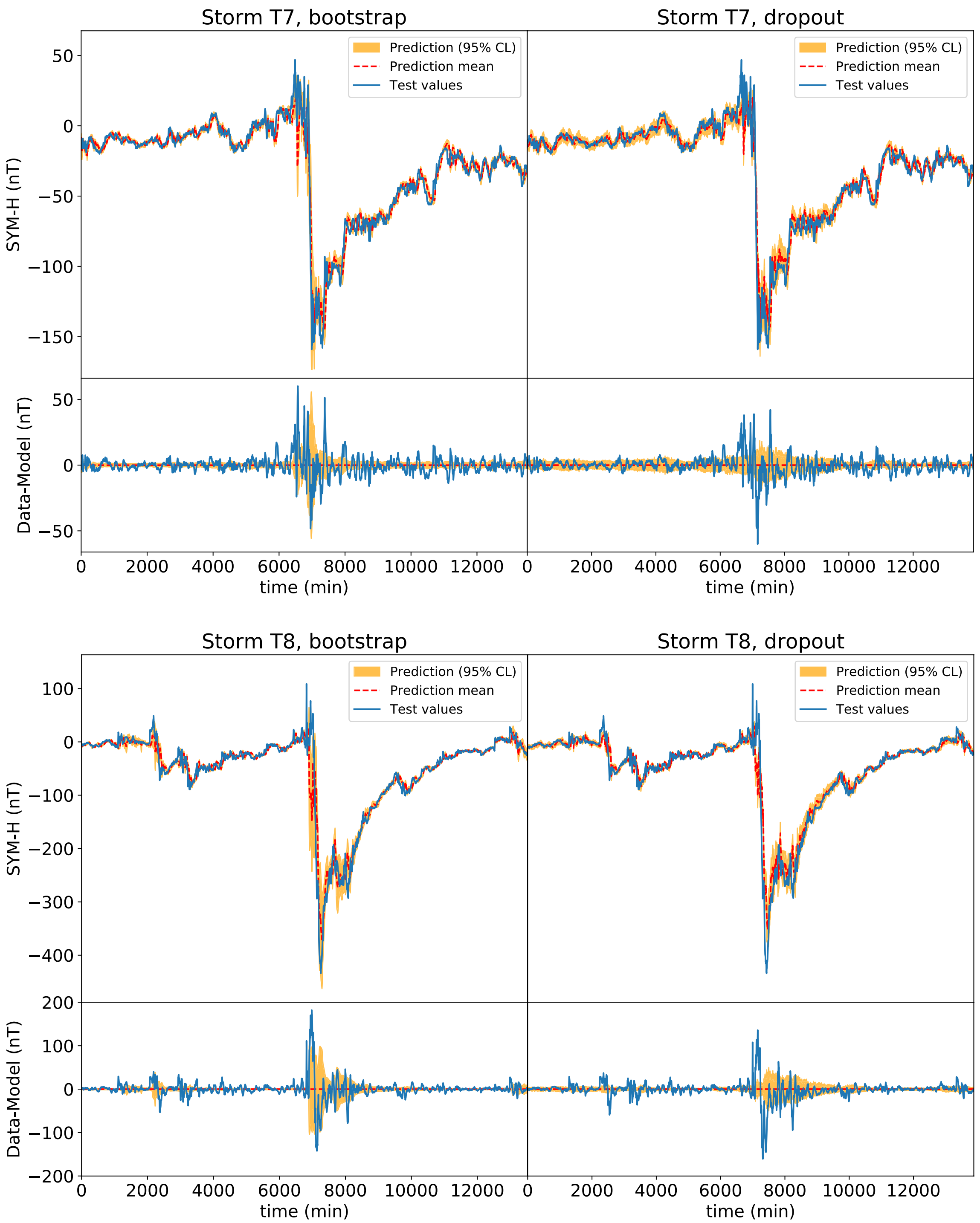

Figure A1. Time-series distributions for all 17 storms in the test sub-data set, showing the 1-hr ahead results using the bootstrap method (left) and the dropout method (right). In all distributions, we show in an orange band the 95% CL (corresponding to 1.96σ), in red dashed line the mean for the one-hour ahead predictions of the SYM-H index from the long short-term memory model, and the test data as a solid blue line. The lower panels represent the residuals with respect to the model prediction mean.

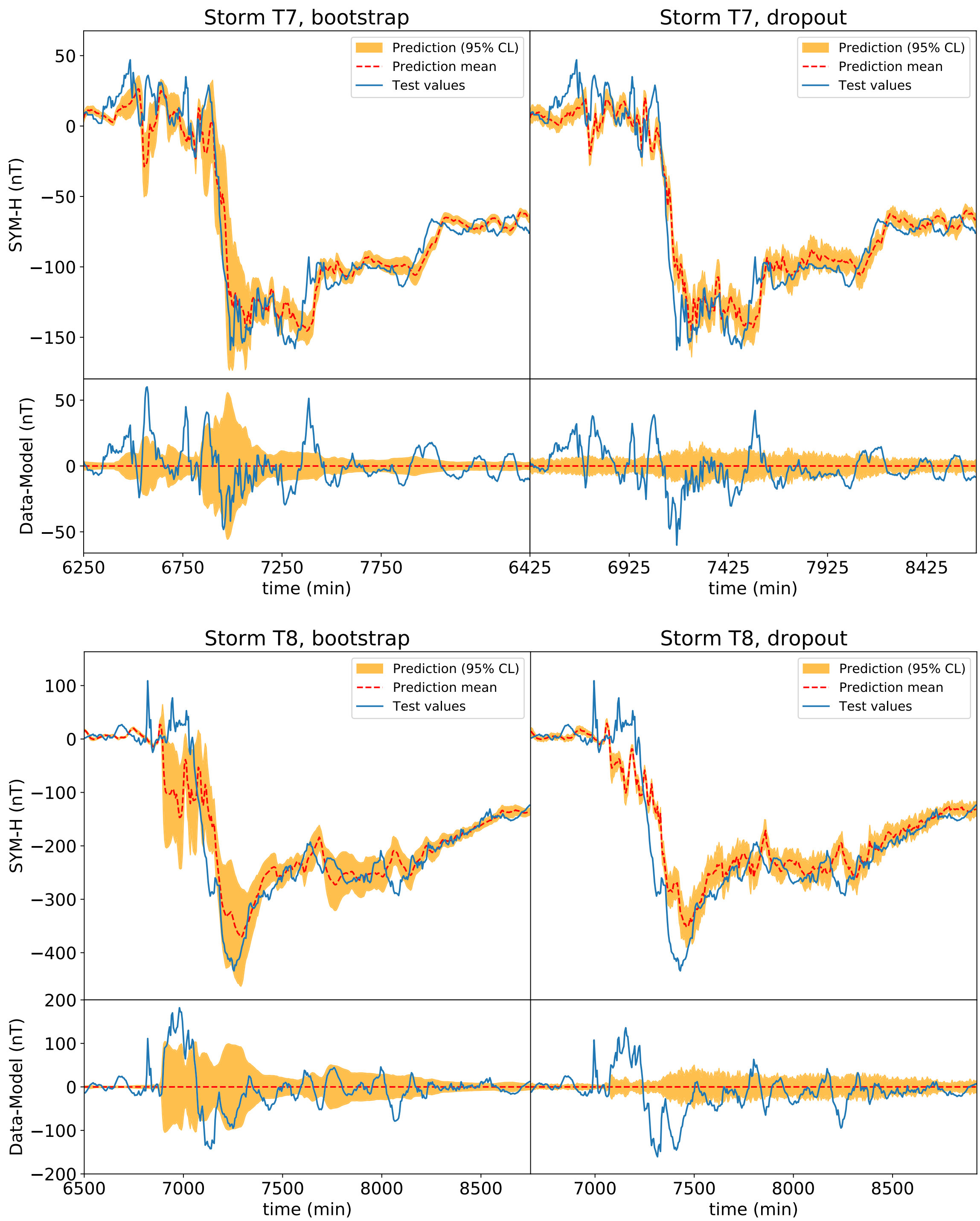

Figure A2. Zoom-in of the time-series distributions around the peaks of maximum activity for storms T7 and T8 in the test sub-data set, showing the 1-hr ahead results using the bootstrap method (left) and the dropout method (right). In these distributions, we show in an orange band the 95% CL (corresponding to 1.96σ), in red dashed line the mean for the one-hour ahead predictions of the SYM-H index from the long short-term memory model, and the test data as a solid blue line. The lower panels represent the residuals with respect to the model prediction mean.

Data Availability Statement

Raw data are obtained from the NASA's OMNIWeb page (https://omniweb.gsfc.nasa.gov). Processed data, high-resolution plots, prediction models (for both bootstrap and dropout) in `h5` format and the scripts involved in the analysis and model building can be downloaded at https://zenodo.org/record/7871494 (SpaceWeather-IFIC, 2023). The list of packages required to run the scripts is listed in the repository, as well, under the file `requirements.txt`; in particular, for building the models, we used `keras==2.3.1` and `tensorflow==2.5.0`. We run them under `NVIDIA GPU CUDA Version: 12.0 Driver Version: 525.85.12`. In addition, the custom `MonteCarloLSTM` layer is built in the script `trainModels.py` in the `lib` folder, and it also allows us to activate the dropout procedure not just in training, but also during data prediction.

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References

Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. https://doi.org/10.48550/arXiv.1907.10902

Bailey, R. L., Leonhardt, R., Möstl, C., Beggan, C., Reiss, M. A., Bhaskar, A., & Weiss, A. J. (2022). Forecasting GICS and geoelectric fields from solar wind data using LSTMS: Application in Austria. *Space Weather*, *20*(3), e2021SW002907. https://doi.org/10.1029/2021SW002907

Bergstra, J., Bardenet, R., Bengio, Y., & Kégl, B. (2011). Algorithms for hyper-parameter optimization. In J. Shawe-Taylor, R. Zemel, P. Bartlett, F. Pereira, & K. Weinberger (Eds.), *Advances in neural information processing systems* (Vol. 24). Curran Associates, Inc.

Bhaskar, A., & Vichare, G. (2019). Forecasting of SYM-H and ASY-H indices for geomagnetic storms of solar cycle 24 including St. Patricks day, 2015 storm using NARX neural network. *Journal of Space Weather and Space Climate*, *9*, A12. https://doi.org/10.1051/swsc/2019007

Boteler, D., Pirjola, R., & Nevanlinna, H. (1998). The effects of geomagnetic disturbances on electrical systems at the earth's surface. *Advances in Space Research*, *22*(1), 17–27. https://doi.org/10.1016/S0273-1177(97)01096-X

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/A%3A1010933404324

Burton, R. K., McPherron, R. L., & Russell, C. T. (1975). An empirical relationship between interplanetary conditions and Dst. *Journal of Geophysical Research*, *80*(31), 4204–4214. https://doi.org/10.1029/JA080i031p04204

Cai, L., Ma, S. Y., & Zhou, Y. L. (2010). Prediction of SYM-H index during large storms by NARX neural network from IMF and solar wind data. *Annales Geophysicae*, *28*(2), 381–393. https://doi.org/10.5194/angeo-28-381-2010

Camporeale, E. (2019). The challenge of machine learning in space weather: Nowcasting and forecasting. *Space Weather*, *17*(8), 1166–1207. https://doi.org/10.1029/2018SW002061

Chakraborty, S., & Morley, S. (2020). Probabilistic prediction of geomagnetic storms and the kp index. *J. Space Weather Space Clim*, *10*, 36. https://doi.org/10.1051/swsc/2020037

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 785–794). ACM. https://doi.org/10.1145/2939672.2939785

Collado-Villaverde, A., Muñoz, P., & Cid, C. (2021). Deep neural networks with convolutional and LSTM layers for SYM-H and ASY-H forecasting. *Space Weather*, *19*(6), e2021SW002748. https://doi.org/10.1029/2021SW002748

Collado-Villaverde, A., Muñoz, P., & Cid, C. (2023). Neural networks for operational SYM-H forecasting using attention and SWICS plasma features. *Space Weather*, *21*(8), e2023SW003485. https://doi.org/10.1029/2023SW003485

Efron, B. (1979). Bootstrap methods: Another look at the Jackknife. *Annals of Statistics*, *7*(1), 1–26. https://doi.org/10.1214/aos/1176344552

Fisher, A., Rudin, C., & Dominici, F. (2019). All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research*, *20*(177), 1–81. Retrieved from http://jmlr.org/papers/v20/18-760.html

Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. *Journal of Machine Learning Research Workshop and Conference Proceedings*, *48*. Retrieved from http://proceedings.mlr.press/v48/gal16.pdf

Gal, Y., Hron, J., & Kendall, A. (2017). Concrete dropout. https://doi.org/10.48550/arXiv.1705.07832

Gruet, M. A., Chandorkar, M., Sicard, A., & Camporeale, E. (2018). Multiple-hour-ahead forecast of the Dst index using a combination of long short-term memory neural network and Gaussian process. *Space Weather*, *16*(11), 1882–1896. https://doi.org/10.1029/2018SW001898

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, *9*(8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735

Iong, D., Chen, Y., Toth, G., Zou, S., Pulkkinen, T., Ren, J., et al. (2022). New findings from explainable SYM-H forecasting using gradient boosting machines. *Space Weather*, *20*(8), e2021SW002928. https://doi.org/10.1029/2021SW002928

Iyemori, T. (1990). Storm-time magnetospheric currents inferred from mid-latitude geomagnetic field variations. *Journal of Geomagnetism and Geoelectricity*, *42*(11), 1249–1265. https://doi.org/10.5636/jgg.42.1249

King, J., & Papitashvili, N. (2005). Solar wind spatial scales in and comparisons of hourly wind and ace plasma and magnetic field data. *Journal of Geophysical Research*, *110*(A2), A02104. https://doi.org/10.1029/2004JA010649

Leontaritis, I. J., & Billings, S. A. (1985). Input-output parametric models for non-linear systems part ii: Stochastic non-linear systems. *International Journal of Control*, *41*(2), 329–344. https://doi.org/10.1080/0020718508961130

Licata, R. J., Mehta, P. M., Tobiska, W. K., & Huzurbazar, S. (2022). Machine-learned HASDM thermospheric mass density model with uncertainty quantification. *Space Weather*, *20*(4), e2021SW002915. https://doi.org/10.1029/2021SW002915

Liemohn, M. W., McCollough, J. P., Jordanova, V. K., Ngwira, C. M., Morley, S. K., Cid, C., et al. (2018). Model evaluation guidelines for geomagnetic index predictions. *Space Weather*, *16*(12), 2079–2102. https://doi.org/10.1029/2018SW002067

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. V. N. Vishwanathan, et al. (Eds.), *Advances in neural information processing systems* (Vol. 30). Curran Associates, Inc. Retrieved from https://proceedings.neurips.cc/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf

Madsen, F. D., Beggan, C. D., & Whaler, K. A. (2022). Forecasting changes of the magnetic field in the United Kingdom from L1 Lagrange solar wind measurements. *Frontiers in Physics*, *10*, 1017781. https://doi.org/10.3389/fphy.2022.1017781

Marsal, S., & Curto, J. (2009). A new approach to the hourly mean computation problem when dealing with missing data. *Earth Planets and Space*, *61*(8), 945–956. https://doi.org/10.1186/BF03352945

Mayaud, P. N. (1980). Introduction. In *Derivation, meaning, and use of geomagnetic indices* (pp. 1–2). American Geophysical Union (AGU). https://doi.org/10.1002/9781118663837.ch1

Mistryukova, L., Plotnikov, A., Khizhik, A., Knyazeva, I., Hushchyn, M., & Derkach, D. (2023). Stokes inversion techniques with neural networks: Analysis of uncertainty in parameter estimation. *Solar Physics*, *296*(8), 406–416. https://doi.org/10.1007/s11207-023-02189-4

Morley, S. K., Welling, D. T., & Woodroffe, J. R. (2018). Perturbed input ensemble modeling with the space weather modeling framework. *Space Weather*, *16*(9), 1330–1347. https://doi.org/10.1029/2018SW002000

Papitashvili, N. E., & King, J. H. (2023a). Omni 1-min data. NASA space physics data facility [Dataset]. NASA. https://doi.org/10.48322/45bb-8792

Papitashvili, N. E., & King, J. H. (2023b). Omni 5-min data. NASA space physics data facility [Dataset]. NASA. https://doi.org/10.48322/gbpg-5r77

Patowary, R., Singh, S., & Bhuyan, K. (2013). A study of seasonal variation of geomagnetic activity. *Research Journal of Physical and Applied Sciences*, *2*, 1–11.

Pinto, V. A., Keesee, A. M., Coughlan, M., Mukundan, R., Johnson, J. W., Ngwira, C. M., & Connor, H. K. (2022). Revisiting the ground magnetic field perturbations challenge: A machine learning perspective. *Frontiers in Astronomy and Space Sciences*, *9*, 869740. https://doi.org/10.3389/fspas.2022.869740

Qin, Z., Denton, R. E., Tsyganenko, N. A., & Wolf, S. (2007). Solar wind parameters for magnetospheric magnetic field modeling. *Space Weather*, *5*(11), S11003. https://doi.org/10.1029/2006SW000296

Raghu, M., Poole, B., Kleinberg, J., Ganguli, S., & Sohl-Dickstein, J. (2017). On the expressive power of deep neural networks.

Rapin, J., & Teytaud, O. (2018). Nevergrad—A gradient-free optimization platform. Retrieved from https://GitHub.com/FacebookResearch/Nevergrad.GitHub

Rasmussen, C. E., & Williams, C. K. I. (2005). *Gaussian processes for machine learning*. The MIT Press. https://doi.org/10.7551/mitpress/3206.001.0001

Rumelhart, D. E., & McClelland, J. L. (1987). Learning internal representations by error propagation. In *Parallel distributed processing: Explorations in the microstructure of cognition: Foundations* (pp. 318–362).

Siciliano, F., Consolini, G., Tozzi, R., Gentili, M., Giannattasio, F., & De Michelis, P. (2021). Forecasting SYM-H index: A comparison between long short-term memory and convolutional neural networks. *Space Weather*, *19*(2), e2020SW002589. https://doi.org/10.1029/2020SW002589

Siddique, T., Mahmud, A. M., Keesee, M. S., Ngwira, C. M., & Connor, H. (2022). A survey of uncertainty quantification in machine learning for space weather prediction. *Geosciences*, *12*(1), 27. https://doi.org/10.3390/geosciences12010027

SpaceWeather-IFIC. (2023). Spaceweather-ific/open_data: v2.0 [Software]. Zenodo. https://doi.org/10.5281/zenodo.7871494

Tan, Y., Hu, Q., Wang, Z., & Zhong, Q. (2018). Geomagnetic index kp forecasting with LSTM. *Space Weather*, *16*(4), 406–416. https://doi.org/10.1002/2017SW001764

Torta, J. M., Marcuello, A., Campanyà, J., Marsal, S., Queralt, P., & Ledo, J. (2017). Improving the modeling of geomagnetically induced currents in Spain. *Space Weather*, *15*(5), 691–703. https://doi.org/10.1002/2017SW001628

Wanliss, J. (2005). Fractal properties of SYM-H during quiet and active times. *Journal of Geophysical Research*, *110*(A3), A03202. https://doi.org/10.1029/2004JA010544

Wanliss, J., & Uritsky, V. (2010). Understanding bursty behavior in midlatitude geomagnetic activity. *Journal of Geophysical Research*, *115*(A3), A03215. https://doi.org/10.1029/2009JA014642

Wanliss, J. A., & Showalter, K. M. (2006). High-resolution global storm index: Dst versus SYM-H. *Journal of Geophysical Research (Space Physics)*, *111*(A2), A02202. https://doi.org/10.1029/2005JA011034

Zhang, W. (1988). Shift-invariant pattern recognition neural network and its optical architecture. In *Proceedings of annual conference of the Japan society of applied physics*.

Zhang, W., Itoh, K., Tanida, J., & Ichioka, Y. (1990). Parallel distributed processing model with local space-invariant interconnections and its optical architecture. *Applied Optics*, *29*(32), 4790–4797. https://doi.org/10.1364/AO.29.004790

Zhelavskaya, I. S., Vasile, R., Shprits, Y. Y., Stolle, C., & Matzka, J. (2019). Systematic analysis of machine learning and feature selection techniques for prediction of the kp index. *Space Weather*, *17*(10), 1461–1486. https://doi.org/10.1029/2019SW002271